EIGSEP Memo 003

Marjum Pass 2026-07: pointing

EIGSEP Team

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Abstract

Where the antenna pointed, and how we know. This memo states the mount model and angle conventions, how the campaign pointing table turns motor, potentiometer, IMU and LIDAR readings into azimuth and elevation, what the 07-17 transmitter raster was, how well its azimuth is known in absolute terms, and how far its elevation axle tilts from the horizontal. Its central tests use the ratio of the transmitter's two orthogonal arms, which depends on the antenna's orientation and on nothing else: the antenna beam and the transmitter drive cancel. Every figure, table and quoted number is generated by `make_products.py` in this directory (Appendix A). Positions and the GPS azimuth reference are in memo 002; the observing configuration and timeline are in memo 001.

1 Summary

- **The mount is a roll mount, not alt-az.** Azimuth turns the antenna about its own boresight; elevation then tips the assembly about the highline. Across 16 transmitter tooth pairs (140.6–199.2 MHz) the arm ratio fits the roll mount with a pooled rms of 0.117 (worst pair 0.198) and the best alt-az variant with 0.294 (best pair 0.242; §2).
- **On 07-17 the potentiometer slipped, not the platform.** Measured step by step from the polarization, the platform stays within 4.2° rms (worst 10.3°) of the commanded (motor) azimuth while the potentiometer drifts to +29.2° from it. The arm ratio fits the commanded azimuth better than the potentiometer-fused azimuth on all 16 of 16 pairs (pooled rms 0.117 against 0.160; §4). The table's `az_deg` has been the commanded azimuth since v2.0.
- **The raster** is 20:26:00–21:28:40 UTC on 07-17: after a 180° slew at 20:25:59, the elevation axis turns continuously through full revolutions while azimuth steps 5.000° once per revolution (every 70.8 s), from 169.35° to 269.36°.
- **The azimuth is left-handed about the boresight.** +motor az turns the azimuth shaft clockwise seen from above: 93 of 94 steps measured by the azimuth IMU on 07-13 to 07-15, median 0.97 per degree. The 07-17 beam fits put the boresight up at el = 0, where a clockwise turn is left-handed. Two photographs across a +15° motor step show the dipole turning that way (§5.1). From pointing table v5 the table negates the motor and potentiometer readings, so its `az_deg` goes into Eq. (1) and `eigsep_base` unchanged.
- **The table's azimuth zero on the raster is +15.8° from the highline** (±0.9° across pairs), not 0°. This comes from the polarization alone, and the HFSS beam fit agrees (§5). With the left-handed azimuth memo 001's transmitter arm labels stand, but arm 0 lies 10.3° from its photographed axis, unexplained.
- **Elevation**: el is the boresight's angle from zenith (0 zenith, +90 horizontal toward bearing 37.84°, 180 nadir). The transmitter is at el = 183.7°. The IMU reads −2.3° from true (§6). Late in the 07-17 raster the box fell up to 33.0° behind the elevation drive on up-legs; the table's elevation, from the IMU, follows the box (§6). The elevation drive failed near 01:41:40 UTC on 07-18, parked at 183.9°, facing the transmitter.
- **The elevation axle is tilted, by an amount that changes at each raise.** The box's IMU reads the axle's tilt out of the horizontal from the gravity component along it: +5.92° over the 07-17 raster, +0.18 ± 0.6° over `fit_v0025`'s 362 bins of the 07-17 night (the error an allowance for the tilt's wander during the night's hold, whose cause is not known), and +2.01° after the re-raise and −0.27° at 30 m (era

means from a settled start). The sensor itself is turned 3.85° in the box (the firmware's calibration: 3.72°), which does not tilt the antenna. The antenna is bolted to the mount on the box, so the axle's tilt is the IMU's. The tilt's sign is taken from the raster beam fit. That fit's own tilt, +4.74°, refitted on pointing table v6, predicts held-out data no better than the IMU's, so the raster cannot separate a change of tilt from a small change of the beam's shape and its fitted value says nothing about the mount. On the night a sky fit rejects the raster's tilt (held-out χ^2 139.8 at it, 27.2 at the IMU's, 27.4 at 0) and is consistent with the IMU's value and with level. Pointing table v6 carries the tilt per sample (§7).

- **Antenna phase center lever arm and flexure**: cross-correlation phase (`box-gnd` × `box-air`) on the transmitter comb through the 07-17 raster sweeps resolves the dipole and balun's 3-D body offset from the elevation axle: $z_{\rm pc} = +55.6 \pm 6.6$ cm along boresight, $x_{\rm pc} = -15.7$ cm along the dipole arm, and $y_{\rm pc} = -0.5$ cm transverse (mean fundamental lever arm 57.5 cm, total $\|\mathbf{r}_{\rm pc}\| = 57.7$ cm; §8). Gravitational flexure adds a 11.3 cm 2nd-harmonic sag when cantilevered horizontally (el = ±90°). Stationary platform wobble is 0.75 cm rms.
- **Table defects** to work around: `EL_POST_FAILURE` is set on every row from 07-14 14:58 onward, including the whole raster (§10).

2 The mount and the angle conventions

A vector in the antenna body frame (boresight $+z$, dipole arm $+x$) maps to local ENU as

$$v_{\rm ENU} = R_z(\psi)\, R_x(\mathrm{el})\, R_z(\mathrm{az})\, v_{\rm body}, \qquad \psi = 142.164^\circ, \tag{1}$$

implemented as `eigsep_base.rotations.mount_rotation`. ψ is the highline direction counter-clockwise from East (bearing 127.84°/307.84°; memo 002). Azimuth is applied first, about the boresight: **it rolls the antenna and changes only the polarization angle; it never moves the beam.** Elevation is applied second, about the highline, so the boresight always stays in the vertical plane through the antenna at bearing 37.84°/217.84°. The alternative $R_z(\psi)R_z(\mathrm{az})R_x(\mathrm{el})$ (tilt, then turn about the vertical) is the alt-az mount; it is not equivalent under any relabelling. The elevation axle is not exactly horizontal: Eq. (2) adds its tilt, which the box's IMU measures (§7).

The antenna and transmitter positions are geometry release `v0005` (memo 002): the release's selected state, the positions the horizon profiles (`curation/horizon_profiles_v0005`) also use; memo 002 quotes posterior means, 0.24 m (antenna) and 0.54 m (transmitter) from them. Its E, N are UTM grid axes; the line of sight is rotated by the 1.52° grid convergence at the antenna into the true-north frame in which ψ and the transmitter's arm bearings are defined.

Why the beam cannot tell the two apart here. The transmitter is 0.40° from the boresight plane and near nadir, so in either mount its angle from the boresight is close to $|180^\circ - \mathrm{el}|$. Total transmitter power is nearly blind to the mount.

Why the polarization can. The transmitter's two arms are orthogonal horizontal dipoles, driven on alternate comb teeth (memo 001). For a tooth on arm 0 and its neighbour on arm 1, 1.95 MHz above, the fraction of their summed power in arm 0 depends only on the angle between the antenna's dipole and the transmitter's arms, projected perpendicular to the line of sight. The antenna beam is common to both and cancels, as does the absolute drive. Under the roll mount the antenna arm's horizontal projection is $R_z(\psi)\,(\cos\mathrm{az},\ \sin\mathrm{az}\cos\mathrm{el})$: near nadir or zenith it rotates fully with azimuth, toward $\mathrm{el} = \pm 90^\circ$ it is squeezed toward the highline, and the sense of rotation reverses between the upper and lower hemispheres. Under alt-az it is $R_z(\psi + \mathrm{az})\,\hat{x}$ at every elevation, so the arm share would depend on azimuth alone.

Test. For each pair we fit the arm-0 share with three free parameters (an azimuth offset, the bearing of transmitter arm 0, and the arm drive ratio) under each mount and each azimuth handedness, using every good sample of the raster (at least 6202 per pair; the beam product's sample, tooth and slew flags applied). Figure 1 shows one pair; Table 1 summarizes all 16. Per pair, the roll mount fits at 0.085–0.198 rms and alt-az at 0.242–0.316: the best alt-az pair is worse than the worst roll pair. Alt-az cannot make the arm share depend on elevation at fixed azimuth, and the data do. The roll-mount residual is largest at the ends of the band (Fig. 2d), where the antenna's own cross-polarization off boresight, which this model leaves out, is largest.

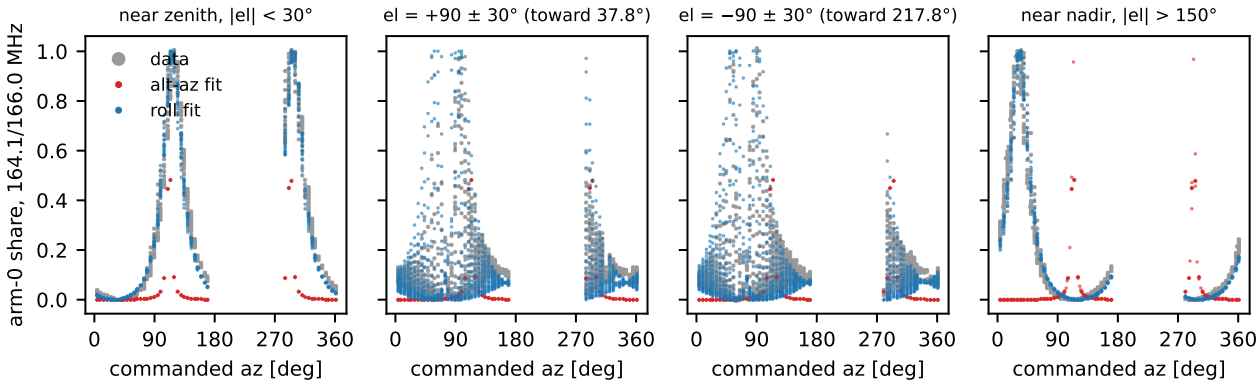


Figure 1: Share of transmitter power in arm 0 against commanded azimuth, in four elevation bands, for one tooth pair (grey), with the best roll-mount (blue) and alt-az (red) fits evaluated at the same samples. The raster covers commanded azimuth 190° through 360° to 91°.

Table 1: Arm-share fits over 16 tooth pairs (140.6–199.2 MHz). The rms pools every residual of every pair, with the worst pair (for alt-az, the best) in brackets; angles are means weighted by sample count, with the range over pairs. "Az zero from highline" is the offset added to the table's azimuth so that az = 0 puts the dipole along the highline; "arm-0 bearing" is the compass bearing of transmitter arm 0, mod 180°. The last row repeats the preferred model on the potentiometer-fused azimuth (§4).

mount	handedness	pooled rms (worst pair)	az zero from highline	arm-0 bearing
roll	left-hand	0.117 (0.198)	+15.8° (+14.2 to +17.0)	175.9° (172.3–179.4)
roll	right-hand	0.119 (0.199)	−16.1° (-17.1 to -14.6)	79.9° (76.5–83.5)
alt-az	left-hand	0.294 (0.242)	—	—
alt-az	right-hand	0.297 (0.246)	—	—
roll, pot-fused az	left-hand	0.160 (0.230)	−4.3° (-4.7 to -1.9)	6.1° (3.5–9.5)

Elevation convention. el is the boresight's angle from zenith by the right-hand rule about the highline pointing west-north-west (bearing 307.84°): 0 is zenith, +90 is horizontal toward 37.84°, 180 (equivalently −180) is nadir. The table reports el in $(-180, 180]$, so the elevation axis turning through nadir shows as a jump from +180 to −180. Beware the wording "180° from zenith" for the transmitter: it sits at el = 183.7°, which the table writes as 183.7° − 360°, not at nadir.

Azimuth convention. az in Eq. (1) is by the right-hand rule about the boresight (about up when el = 0), so increasing az turns the dipole counter-clockwise seen from above at el = 0, and clockwise seen from above when the antenna faces down. The hardware turns the other way: +motor az is left-handed about the boresight (§5.1). Pointing table v5 therefore negates the motor and potentiometer readings, `az_deg` = −(`motor_az_deg` + `az_commanded_offset_deg`) on commanded rows, and Eq. (1) and `eigsep_base` are unchanged. Every azimuth in this memo is the table's. At az = 0 the dipole lies along the highline *by definition*. Whether the table's az = 0 meets that definition is the question of §5.

3 The sensors and how the table combines them

The pointing table (`marjum-2026-07/curation/pointing_table.parquet`, v5, 1,227,956 rows from 5,120 files) has one row per correlator sample, joined on `t_utc`. Its README documents the columns, the clock validation, and coverage by era; this section records the rules and what is known about each sensor. The streams are in each correlator file's `metadata` group, natively aligned with the spectra.

Motor counts (az and el, 180/11300 deg per step). Smooth and finely quantised, but relative: they re-home with jumps of up to ∼46°, and they say where the drive was told to go, not where the platform went.

Potentiometer (az). The only absolute azimuth sensor. Scatter 1.73° on a static platform, and it slips: through the 07-17 raster it drifted from +2.6° to +29.2° against the platform (§4).

IMU elevation. Gravity-referenced and absolute, scatter 0.24°; read −2.3° from true (§6).

IMU azimuth. Dead from 07-16 on, and drifting at ∼190°/hr where alive. Not used.

LIDAR. A rangefinder mounted 90° from the boresight; it points at nadir when el = +90. 46% of its 52,742 readings are the 250 m out-of-range sentinel. It is a geometry check, not an altimeter (§6, §9).

Elevation is the IMU wherever it is finite. Where it is not, the motor fills in, flagged `EL_NO_IMU` and `suspect`. Where the elevation moves faster than 20°/s between adjacent samples, both samples are flagged `EL_SOLUTION_GLITCH` and `suspect`, since the rate does not say which one is wrong. On the raster 146 rows are flagged, near el = ±180°, among them 60 onboard-solver jumps of ±120° or 180° in `imu_el_deg`; every such jump is flagged.

Azimuth (`az_deg`, v2.0) is the *commanded* azimuth: motor counts give the motion, and the potentiometer sets one absolute offset per anchor segment; from v5 the sum is negated (§2, Azimuth convention). The raw columns `motor_az_deg` and `pot_az_deg` keep the readings' sense. A new segment starts at every re-home (a count jump > 10° between samples), every telemetry gap > 5 s, and every big slew (continuous motion spanning > 20°; the raster's 5° steps never qualify). The offset is the median of pot−motor over the first 10 minutes of the segment after its slew. There are 165 segments campaign-wide. The rule rests on two facts: the az and el drives never run at the same time and are locked when idle (Aaron), and on 07-17 the platform demonstrably followed the motor (§4). The v1.x azimuth (a complementary filter re-anchoring the motor to a ±30 s rolling median of the potentiometer) survives as `az_fused_deg`. Code: `data-analysis/notebooks/arp/marjum-2026-07/pointing/fuse.py` (`commanded_azimuth`, `fuse_azimuth`, `fuse_elevation`).

What commanded azimuth assumes. That the platform follows the motor between big slews. That is tested only on the 07-17 raster. Absolute azimuth comes from one potentiometer reading per segment, so it is no better than the potentiometer's zero at that moment, and segments either side of a slew can differ by whatever the platform slipped during it.

Height is the era's antenna height above the DEM surface, from `curation/horizon_profiles_v0005` (memo 002, Heights), on every row with a height era; the LIDAR range no longer overrides it (from v4).

Table 2: Flag bits in pointing table v5: rows set campaign-wide, and the share of the raster (20:26:00–21:28:40) on which each is set. `AZ_SLIP_EVENT` and `AZ_SLIP_RAMP` now mark potentiometer/motor divergence, not platform slip.

bit	name	rows, campaign	share of raster
1	NO_METADATA	442,080	0%
2	AZ_NO_POT	458,800	0%
4	AZ_SLIP_EVENT	17,701	1%
8	EL_NO_IMU	485,396	0%
16	EL_STUCK	4,410	0%
32	UNCOMMANDED_MOTION	9,560	0%
64	NO_ESTIMATE	464,359	0%
128	HEIGHT_ASSUMED	1,227,956	100%
256	EL_POST_FAILURE	700,556	100%
512	AZ_SLIP_RAMP	27,876	20%
1024	EL_SOLUTION_GLITCH	983	2%
2048	AZ_POT_DIVERGES	26,499	71%
4096	AZ_SLEW	3,677	1%
8192	AZ_FUSED_FALLBACK	8,911	0%

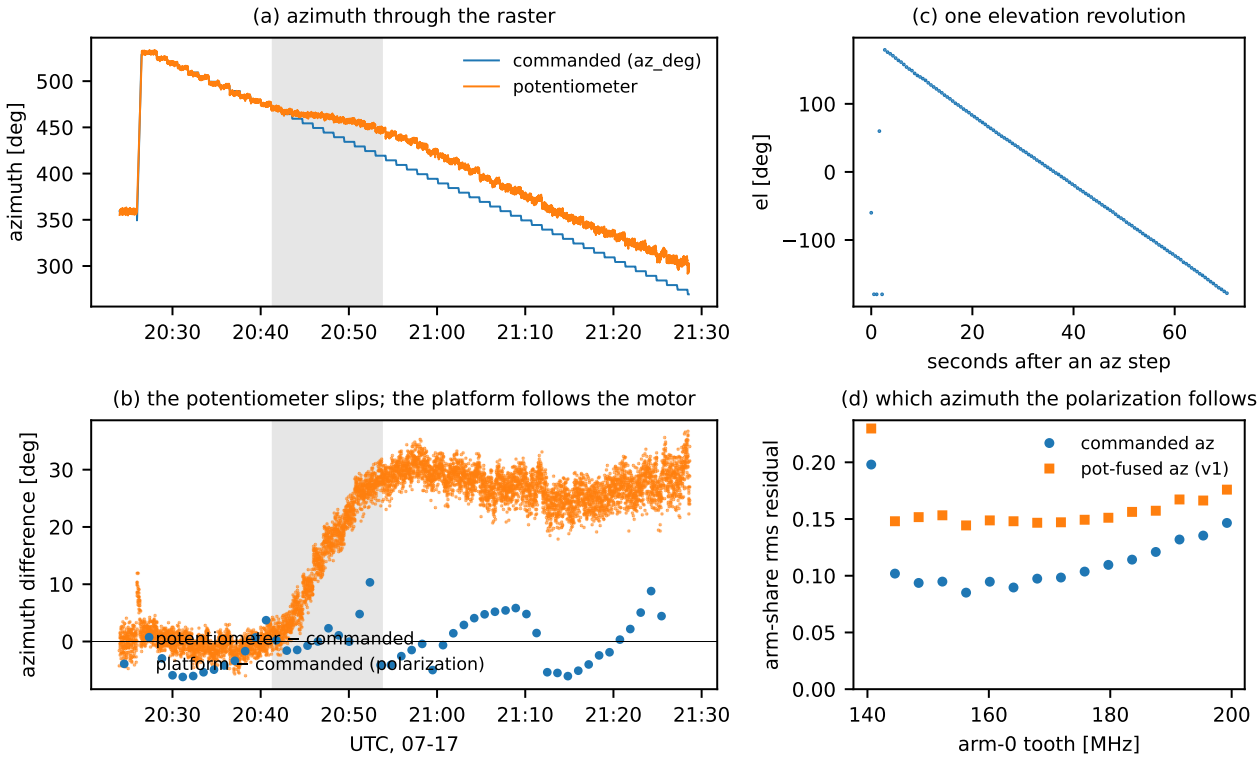


Figure 2: (a) Commanded and potentiometer azimuth through the raster. (b) Their difference; the shading is the interval v1 of the table treated as a 27° platform slip. (c) Elevation through one revolution between azimuth steps. (d) The arm-share residual of the roll-mount fit (§2) per tooth pair, using commanded azimuth and using the potentiometer-fused v1 azimuth.

4 The 07-17 transmitter raster

The motor's azimuth record fixes the raster's structure exactly (Fig. 2). At 20:25:59 the azimuth slews by 180° to 169.35°. From 20:28:12 to 21:28:23 it then steps 52 times, every step 5.000° of motor az on average (4.986–5.002, the motor quantisation), one per elevation revolution, at a mean interval of 70.8 s, ending at 269.36°: 260.0° in all. The table's az, being left-handed, decreases by each step. Between steps the elevation axis turns continuously through a full revolution, so each revolution passes the transmitter once near el = 183.7°. The beam fits use 20:26:00–21:28:40. Read from the potentiometer, the same motion looks like uneven steps of ∼6° ending near 55°, which is how it was first described.

The potentiometer slipped, not the platform. The potentiometer does not track the motor (Fig. 2b): their difference goes from +2.6° at the start to +29.2° at the end, changing fastest (154°/hr) over 20:41–20:54, which v1 of the table recorded as a 27.35° platform slip. Either the platform lagged the motor or the potentiometer lagged the platform. The polarization decides: the arm share follows the commanded azimuth (pooled rms 0.117) better than the potentiometer-fused azimuth (0.160) on every pair (Fig. 2d).

The direct test is to measure the platform's azimuth from the polarization one step at a time. Holding each pair's fitted arm bearing, drive ratio and offset fixed, we fit one extra azimuth shift per raster step to all pairs together (Fig. 2b, blue). Over 50 steps the shift has rms 4.2° and never exceeds 10.3°; inside 20:41–20:54, while the potentiometer reaches +28.7° from it, the median |shift| over its 11 steps is 1.5°, no larger than the 4.1° median over the other steps. The largest single-step shift in that window is not quoted: it changes by several degrees under sub-degree changes in the assumed antenna and transmitter positions. **The platform followed the motor.** The smooth few-degree wander of the shift is a bound on platform-motor differences and model error together.

By time, pooled over pairs:

pooled arm-share rms	before 20:41:24	20:41:24–20:53:43	after
commanded azimuth	0.073	0.179	0.102
potentiometer-fused azimuth	0.141	0.205	0.148

Both azimuths fit worse inside 20:41–20:54 and the minutes after. Since the per-step shift stays small there, the extra scatter is not pointing; its cause is not identified. This agrees with the beam-model test that led to v2.0 (data-analysis `e0ffa78`, `notebooks/arp/marjum-2026-07/single_ch_beam_fit.ipynb`): the unfitted HFSS beam's held-out test residual at 156.25 MHz is 0.071 with commanded azimuth against 0.197 with v1. The present test does not use the beam model at all.

5 Absolute azimuth

Memo 002 places az = 0 (dipole along the highline) at bearing 37.84° for the boresight plane, from the GPS anchors, ±0.6°; that is the geometry of the highline and is not in question. What is in question is whether the table's `az_deg` = 0 puts the dipole along the highline. It inherits the potentiometer's zero, one reading per segment, and the potentiometer slips.

5.1 Azimuth handedness

Eq. (1) turns the body right-handedly about the boresight (body $+z$, the $+z$ of the HFSS beam), with el = 0 at zenith. At el = 0, then, +az turns the dipole counter-clockwise seen from above. The polarization fits cannot tell which way the hardware turns (§5.2). It depends on two physical facts: which way the azimuth shaft turns for +motor az, and which way the boresight faces at el = 0. They have to be measured separately, because turning the body 180° about the dipole arm maps $R_x(\mathrm{el})R_z(\pm\mathrm{az})$ to $R_x(\mathrm{el}+180°)R_z(\mp\mathrm{az})$: a reversed shaft and a reversed boresight leave the dipole arm in the same place. `imu_az` measures the first and the beam the second (Table 3). Both come from `validate_az_handedness.py`, run at commit 320e242. Two photographs then show the turn directly, without the IMU.

The shaft turns clockwise seen from above. `imu_az`, a BNO085 accelerometer, rides the azimuth shaft. It reported from 07-13 02:43 to 07-15 17:09 UTC, with box-air on the ground and the elevation drive parked at motor el = 0 for every azimuth move (model el −3.1°, range −4.3 to 0.0°, from `imu_el` in the pointing table). Gravity then sits 4.4–4.6° off the shaft, so the specific force traces a small circle in `imu_az`'s frame as the shaft turns. A sensor turned counter-clockwise about up by θ sees any fixed vector turn by $-\theta$. So the circle's phase about up changes by −1 per degree of motor az if the shaft turns counter-clockwise from above, and by +1 if it turns clockwise.

"Up" in `imu_az`'s frame comes from the 2026-07-09 picohost calibration that every file carries in `header/imu_calibration`: a full elevation revolution (13 stops) with the azimuth parked. That turn tips the shaft through every direction relative to gravity, so the fitted bias ($5.1\,\mathrm{m\,s^{-2}}$) includes its component along the shaft. An azimuth turn alone cannot do this: it leaves every reading on one circle, and a bias along the shaft cannot then be told from gravity along it. The calibration still held during the test. With its bias removed the reading's magnitude is 0.992 of the calibrated g (5–95%: 0.978–1.006), and the specific force's component along the calibrated up is at least 0.969 at every settled azimuth. The shaft axis is 9.5° from the calibration's up, farther than gravity is from the shaft, so the phase is measured about the fitted circle centre. About the calibrated up itself the slope is +0.08, which is meaningless because the origin lies outside the circle.

Between the 902 settled azimuths (motor at its target for at least five samples) there are 103 steps of 1–150°: 54 with +az and 49 with −az. In 9 of them the motor count changed while the phase moved by less than 1°, so the shaft did not turn; these are count re-homes. Of the other 94, 93 turn clockwise seen from above (Fig. 3b), with a median phase change of 0.97 per degree of motor az (5–95%: 0.33–1.20). The same code, run on a simulated sensor turned counter-clockwise, returns −1. So **+motor az turns the azimuth shaft clockwise seen from above**. This rests on four things outside this test:

- the BNO085's axes are right-handed (its datasheet, §4);
- its accelerometer reports specific force, $+g$ on the upward axis at rest;
- the firmware (`pico-firmware imu.c`) passes the axes through unchanged, checked on its public `main`, not the build that flew;

- the azimuth motor's direction did not change between these days and the 07-17 raster (Aaron, 2026-10-26). The readings agree: potentiometer and motor move together on each of the 6 days 07-13 to 07-18 (slopes +0.50 to +1.02), and every potentiometer calibration in the campaign's 4,965 correlator files (15 distinct settings) has a positive slope (303–324 deg/V).

The boresight faces up at el = 0. The HFSS pattern is not front–back symmetric, so the 07-17 transmitter raster can tell the boresight up from down. We fit it with each of the four combinations in Table 3: the az and el zeros free, the transmitter and antenna at memo 002's `v0005` posterior means, on 6552 samples and 64 comb teeth, scored on held-out azimuth stripes. The raster is loaded and flagged as `data-analysis/scripts/marjum-2026-07/fit_beam.py` does it, on pointing table v5, and fitted with `eigsep_data.beam_mapping` (466c0d1). With the boresight up the held-out rms is 0.093 (left-handed) and 0.103 (right-handed). With it down the fits are 2.9 times worse: 0.274 and 0.267. The beam fixes the boresight. Its preference between the handednesses, 0.011 in rms, is small beside that, and is not used here.

Two photographs show the turn. IMG_2217 and IMG_2226, from memo 002's 91 m parallax set, show the bowtie face-on from 111 and 119 m during the 07-17 raster (Fig. 4). Between them the motor az went from −90 to −75° in steps that were all positive (table az 79.4 to 64.3°). The elevation was sweeping continuously, so each photograph is taken at the moment the pointing table's el crosses 90° (mast horizontal), the crossing nearest before its EXIF time (20:49:22.6 and 20:52:14.9 UTC). Both photographs show the mast pointing near the camera, which only happens near that crossing. The crossings fall 12 and 8 s before the EXIF times, as if the phone clock ran ahead; the 4 s difference between the two is about 20° of elevation, so the photographs fix the elevation only that well. At el = 90° the mast is 33° from the line to either camera, so both look at the dipole from in front of the mast.

Through memo 002's `v0005` camera poses, with each handedness's az zero from Table 3's beam fits, left-handed puts the upper arm tip at −6° from the image vertical in IMG_2217 and +9° in IMG_2226; right-handed puts it at +3° and then −8°. Positive angles are toward image right, which is 335° true. The two handednesses predict opposite turns. In the photographs the blades are close to vertical in IMG_2217 and the upper blade leans toward image right in IMG_2226, the left-handed turn (Aaron's reading of the photographs, checked here against the pointing). The lean looks larger than the predicted change. The elevation's uncertainty and the mast's 33° angle to the line of sight both enter, so the photographs are used for the sense of the turn only. They use the beam only for which way the mast faces, and no IMU.

Together: the azimuth is left-handed. With the boresight up at el = 0, a clockwise turn from above is a left-handed turn about the boresight. It is the only combination consistent with both measurements, and the photographs show the same turn. Pointing table v5 adopts it by negating the readings: on the raster its az changes by −1.00 per degree of motor az (173 steps), and Eq. (1) holds for it as written.

Table 3: The four combinations of azimuth sense and boresight direction at el = 0. The third column is the turn of the azimuth shaft, seen from above, that each predicts for +motor az; `imu_az` measures clockwise. The last two columns are the 07-17 raster fits' median normalized residual rms on the training samples and on held-out azimuth stripes (`fit_beam.py` az_el model; HFSS beam; `v0005` positions; pointing table v5).

az sense about the boresight	boresight at el = 0	shaft turn for +motor az, from above	imu_az (§5.1)	beam fit rms train	beam fit rms held out
left-handed	up	clockwise	consistent	0.094	0.093
left-handed	down	counter-clockwise	excluded	0.264	0.274
right-handed	up	counter-clockwise	excluded	0.103	0.103
right-handed	down	clockwise	consistent	0.256	0.267

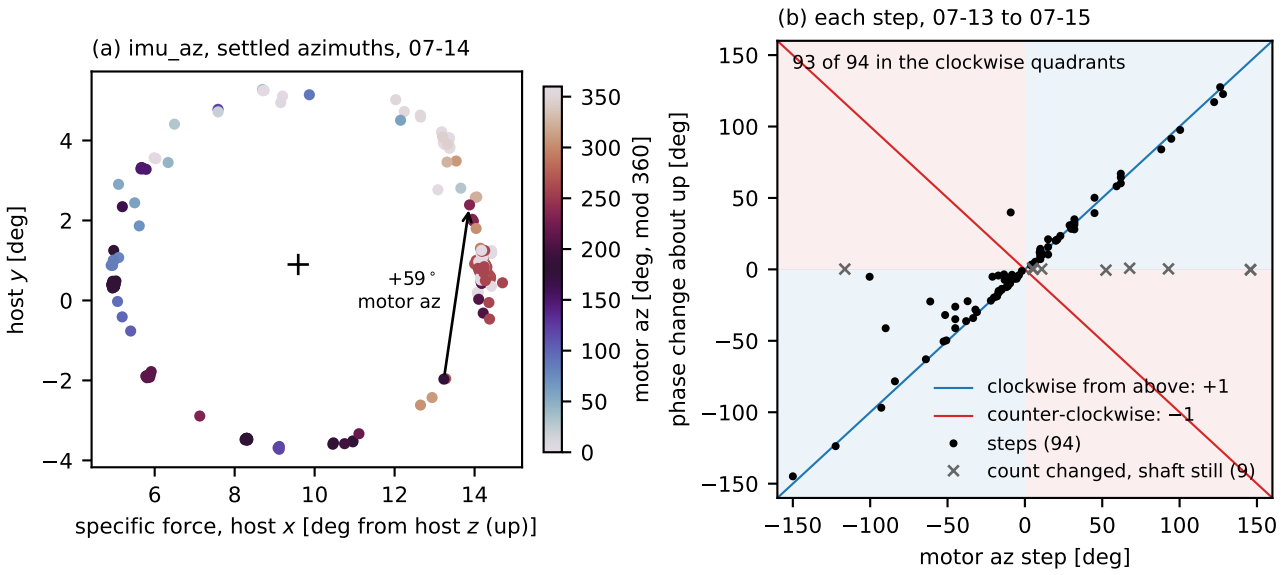


Figure 3: The azimuth shaft's turn from `imu_az`, read from the raw correlator files' metadata with the 2026-07-09 calibration's bias and frame. (a) The specific force at each settled azimuth on 07-14, as angles from the calibration's up, coloured by motor az. The cross is the fitted circle centre, the shaft axis. The view is from above (host z toward the reader). The arrow is one +motor az step: in the sensor's frame the specific force moves counter-clockwise about the centre, so the sensor, and the shaft, turned clockwise. (b) The phase change about the circle centre against the motor az step, for every step from 07-13 to 07-15 (the centre is fitted per day). Blue quadrants: the shaft turns clockwise seen from above (left-handed); red: counter-clockwise (right-handed). Crosses are count changes during which the shaft did not turn.

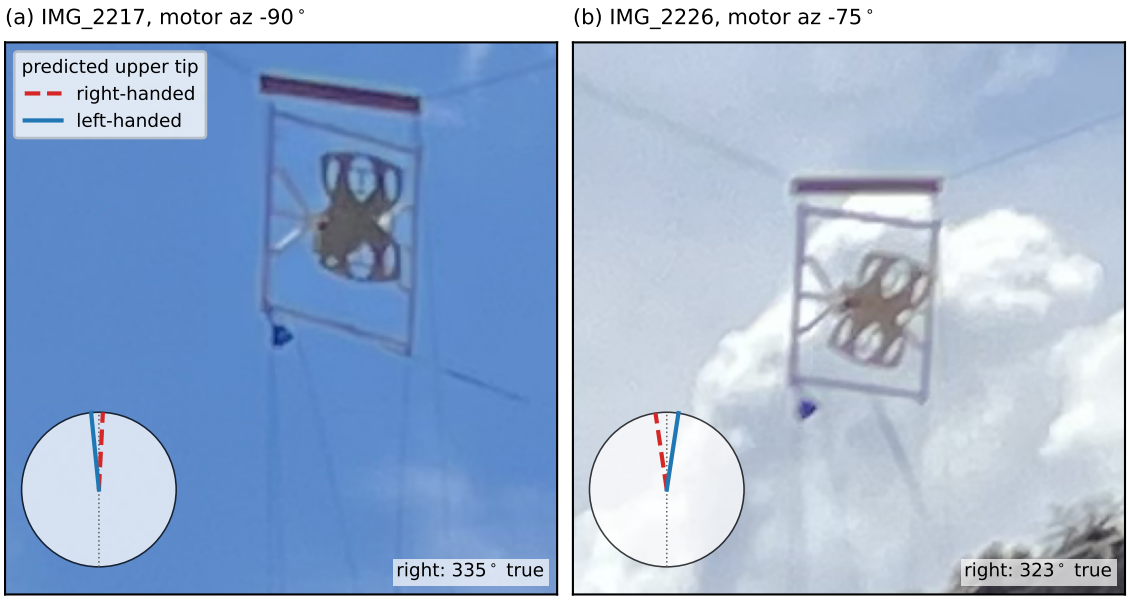


Figure 4: The dipole seen nearly end-on, from in front of the mast, before and after the +15° motor az step: crops of IMG_2217 and IMG_2226, centred on their labelled antenna point and upright. Each dial shows, on the same image axes, the direction of the dipole's upper tip that each handedness predicts at the el = 90° crossing (§5.1): memo 002's `v0005` camera poses and the release's antenna position, with the az zeros of the beam fits in Table 3. The dotted line is image vertical. "Right" is the true bearing of image right. The predictions are model-only and are not drawn on the dipole itself.

5.2 The zero

The polarization measures it. The roll-mount fit's azimuth offset is not degenerate with the transmitter's arm bearing, because the cos el factor in the arm's projection is referenced to the highline, which is fixed. It gives +15.8° (±0.9°; +14.2 to +17.0 across pairs) on the raster's segment, and puts transmitter arm 0 at bearing 175.9° (172.3–179.4). The HFSS beam fit on the same table (`derived/beam/empirical_raster_v0017`) independently wants +15.9° (`table_az_zero_from_highline_deg`) and $\alpha \approx 8°$, an arm-0 field along bearing 172° (memo 001).

Free fits cannot choose the handedness. Reversing it (az → −az in Eq. (1)) fits as well (0.119 against 0.117; Table 1), with the offset of the opposite sign (−16.1°) and arm 0 mirrored about the highline, to 79.9°. Each solution is the mirror image of the other, which is why §5.1 measures the handedness without the polarization.

The photographs constrain it. Memo 002 measures the transmitter's rx6 axis at 165.6° true from the feed's corners in IMG_2203 (sd 0.83° over the sampled v0005 poses; `derived/tx_feed_orientation/v0003`), and rx1 at 75.6°. With arm 0's bearing held at one of these, the polarization fixes the table azimuth at which the antenna's dipole lines up with each arm, and the handedness then fixes which way the dipole turns from az = 0 to get there: the two handednesses imply different zeros, and the off-nadir elevations test each one independently of the bearing. Table 4 gives all four combinations.

Table 4: Arm-share fits with transmitter arm 0's bearing held at the photographed rx6 or rx1 axis (memo 002), for each azimuth handedness; only the azimuth offset and the drive ratio are free. The free fits of Table 1 are repeated for reference.

arm 0 is	handedness	az zero from highline	pooled rms (worst pair)
rx6 (165.6°)	right-hand	+75.8° (+72.6 to +77.5)	0.217 (0.258)
rx6 (165.6°)	left-hand	+19.5° (+17.3 to +23.1)	0.172 (0.246)
rx1 (75.6°)	right-hand	−15.0° (-17.0 to -12.3)	0.132 (0.205)
rx1 (75.6°)	left-hand	−69.4° (-72.5 to -63.9)	0.245 (0.286)
free	left-hand	+15.8°	0.117
free	right-hand	−16.1°	0.119

Only one combination fits nearly as well as the free fit: **right-handed azimuth, with arm 0 along rx1** (rms 0.132, against 0.117 free; better than the next-best combination on 15 of 16 pairs), giving a zero of −15.0°. Arm 0 along rx6 requires the left-handed azimuth, with a zero of +19.5° and a clearly worse fit (0.172): the data put arm 0 10.3° from the photographed rx6 axis, and the zero cannot move to absorb it. The other two combinations imply zeros of +75.8° and −69.4° and fit worst.

An arm gap is open. Three statements cannot all hold: the photographed arm geometry, the handedness of §5.1, and memo 001's identification of arm 0 (teeth on channels ≡ 0 mod 16) as rx6. The first version of this memo assumed a right-handed azimuth and inferred a swapped arm label somewhere in the chain behind memo 001's identification. With the measured left-handed azimuth the labels stand as photographed (arm 0 = rx6), the table's zero on the raster is +15.8° (+19.5° with arm 0 held at rx6), and the fit puts arm 0 10.3° from the photographed rx6 axis, which nothing here explains (§11). Either way, **the table's zero on the raster is ∼15–19° from the highline, not 0**.

Other epochs. The 16° applies to the raster's anchor segment only. The LIDAR terrain fit (memo 002, Azimuth zero) uses the 07-18 01:37 sweep, a different segment and a different potentiometer state, and at the `v0005` antenna finds the zero 1.3° ± 0.8° from the GPS highline value (`derived/lidar_constraint/v0003`); nothing requires the two to agree, given a potentiometer that slipped ∼27° in an hour. For other segments, no absolute azimuth measurement exists, and their `az_deg` should be read as "potentiometer zero, to within tens of degrees".

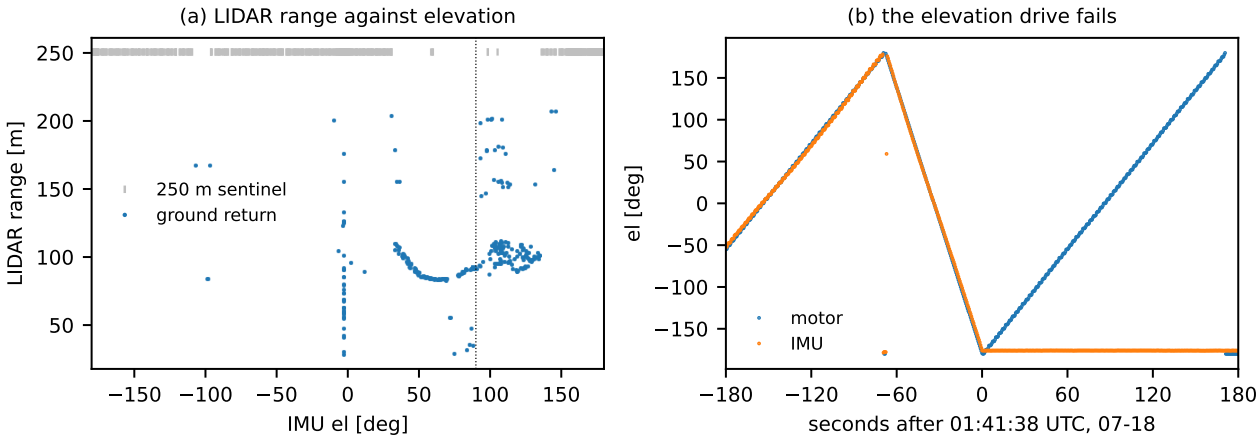


Figure 5: (a) LIDAR range against IMU elevation for every LIDAR reading in the campaign. Ground returns span el 0–150°, with the long dwell at el ≈ 88°, where the LIDAR points at nadir; facing up it reads the 250 m sentinel. (b) Motor and IMU elevation around the 07-18 failure: the motor keeps sweeping while the IMU stays fixed.

6 Elevation: zero, the LIDAR, under-travel, and the failure

Zero. The LIDAR is mounted 90° from the boresight and points at nadir when el = +90. Its 492 returns at 85–100 m within 10° of el = +90 (the long dwell at the final height) have mean range 92.12 m and mean IMU elevation 87.7° (Fig. 5a). If the LIDAR is exactly 90° from the boresight, the IMU reads −2.3° from true (about −2.3° with the boresight at zenith). That angle is not measured, so this route gives the IMU zero plus the LIDAR's mounting error. The beam fit on pointing table v5 (`v0017`; memo 005's reference is now v0018, on pointing table v6) holds the antenna and transmitter at memo 002's `v0005` posterior means and wants only +0.05° near the transmitter, with a tilt of the elevation axle of +4.74° (§7). The two routes are not yet reconciled (§11). The *sign* and origin of el are not in doubt: the LIDAR sees ground with el between 0 and 180 and almost never facing up (el near −90), the right-hand rule about the west-pointing highline gives the same, and the corrected LIDAR geometry puts nadir at el = +90 (legacy MEMO-012 §4.1, in git history before this memo).

Under-travel late in the raster. From leg 45 on, the box turns less than the elevation drive's step count on up-legs (Fig. 6): the counts advance at the normal rate, but the box lags them. It does not stall, except momentarily on leg 45. The counts are open-loop (§3), so they cannot tell steps the motor missed from a slip between the motor and the box. The drive angle is `motor_el_deg`, unwrapped in time; the raster's 54 legs run between its reversals. The residual is `imu_el_deg` minus the drive angle, with the constant offset between their zeros subtracted (−2.03°, the median over legs 0–44), and the 60 onboard-solver jumps removed (§3). Changes are taken in each leg's direction of travel, so a box that turns less than the drive gives a negative change on an up-leg and a positive one on a down-leg. Through leg 44 the residual's leg medians stay within ±4.1°. Afterwards the loss is concentrated at drive angles 90–150°: across that window up-legs 45, 47, 49 and 51 lose 13.3, 4.5, 5.6 and 5.7° more than the median of the 22 earlier up-legs, or 12.1, 4.1, 5.1 and 5.1 times their standard deviation. Over the rest of the sweep their excess is −2.4 to +1.6°, within 1.3 standard deviations; leg 45 loses part of it before the window and regains part after (Fig. 6b). Down-legs show no comparable loss (Fig. 6c): across the same window the late down-legs change by +0.2 to +1.7° more than the earlier ones, at most 1.2 standard deviations. Nor do they give the up-legs' loss back: their net change over a sweep, +1.3 to +3.9°, is within 1.3 standard deviations of the earlier down-legs' +1.5 ± 1.9°, where a recovery would raise it by the up-legs' loss. Leg 45 nearly stalls 3 times: in single 0.54 s integrations the drive advances 2.9° while the box moves −0.3 to +0.6°, which is 58% of its window loss. On the later legs the box keeps moving. No integration loses more than 1.4°, and over the 34–36° of drive angle in which 10–90% of the loss accrues, the box turns 10–13% less than the drive, relative to the earlier legs. The window was found on these legs, so a trials factor applies: scanning 60° windows along each earlier up-leg, 2 of the 22 reach leg 47's z somewhere,

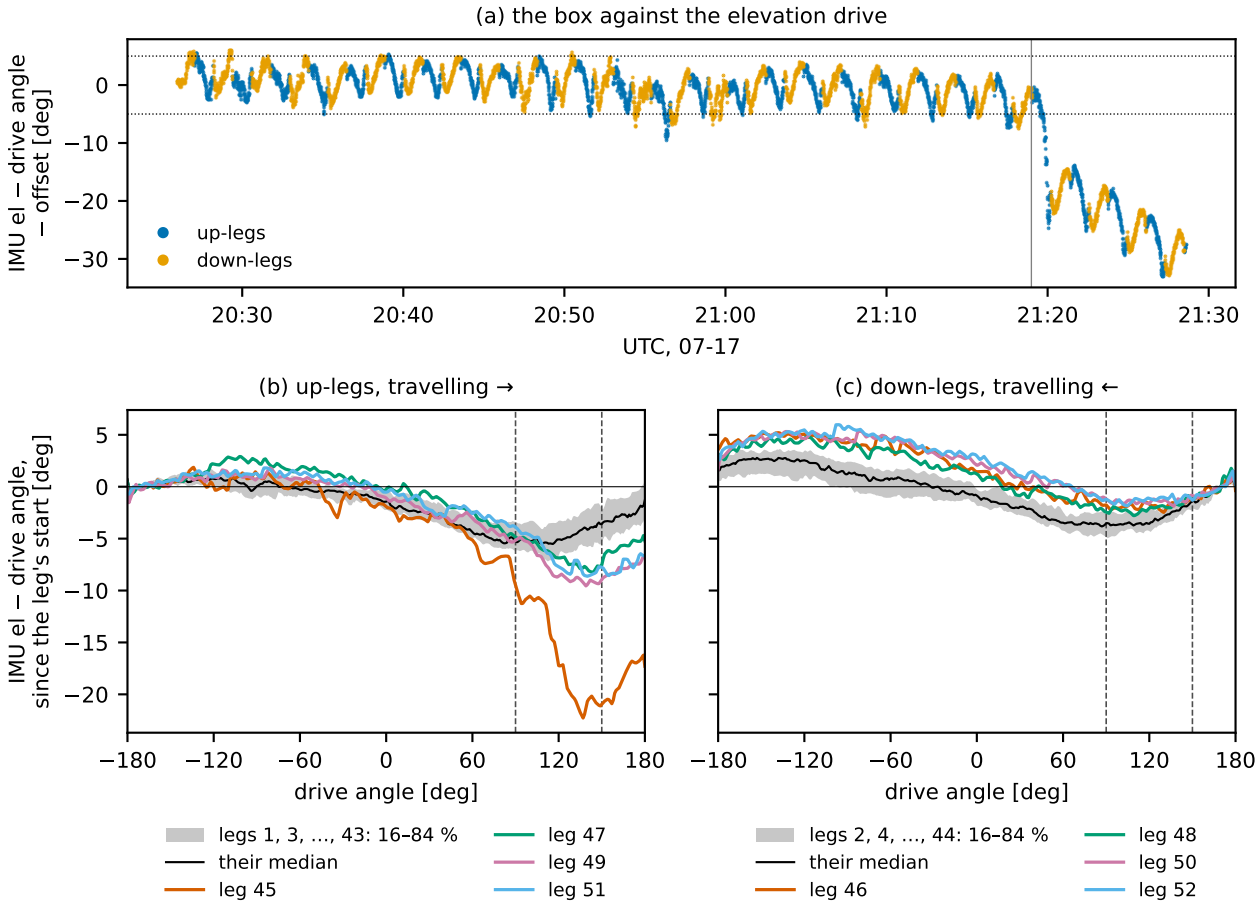


Figure 6: The box against the elevation drive through the 07-17 raster (20:26:00–21:28:40 UTC, pointing table v5; the 60 solver-jump rows removed). (a) The box's elevation from the IMU (`imu_el_deg`) minus the drive angle from the step counts (`motor_el_deg`, unwrapped), on every row, coloured by the leg's direction of travel. The constant offset between the two zeros, −2.03° (the median over legs 0–44, where the box follows the drive), is subtracted, so 0 means the box is where the counts put it and negative values mean it is behind. Dotted lines at ±5°; the grey line marks the start of leg 45. From there the box falls behind on up-legs and keeps the lag on down-legs. (b) Along each up-leg, which travels left to right: the same quantity against drive angle, relative to its level where the leg starts (its median over the first 20°), so a falling curve means the box is falling further behind the counts. Grey: the 22 earlier up-legs (16–84 % range, and median). Even healthy legs are not flat, so the late legs are compared with them, not with zero. Colours: the late up-legs. Dashed lines: the 90–150° window over which the text's numbers are computed. The late up-legs drop away from the band between them, and how far they drop below the band's own change across the window is the excess quoted in the text. (c) The same for down-legs, which travel right to left from +180°: the late down-legs keep the earlier legs' shape, with no comparable loss in the window (§6).

and none reaches those of the other three. By leg 52, the last full leg, the box lags the drive by 28.5° (at most 33.0°, on leg 51), and 1,295 of 6,930 raster rows (18.7%) are more than 5° off.

The table's elevation is the IMU's, so these rows are pointed correctly; `motor_el_deg` is not a pointing on them, and the raster is not a uniform sweep there. The cause is not established. A mechanical bind, a loss of drive torque, an azimuth-dependent load and supply-voltage sag remain unresolved possibilities.

What the current monitor shows. The shared supply current detects the elevation drive: on every late leg it draws 0.24–0.29 A above the nearby parked background, so the drive remains energised during the under-travel. The pooled running-minus-parked estimates are 0.273 A on early legs and 0.264 A on late legs. With both axes parked, the background stays near 2.493 A: its fitted change over the raster is $+7.2 \pm 3.9$ mA, and the 5 parked samples in the final 15 minutes depart from the earlier trend by only $+2.7 \pm 4.2$ mA. These are no detected changes; the sparse parked samples do not bound brief excursions. After background subtraction, the downward sweeps' draw has a gradual trend of -36 ± 4 mA/h, but neither direction shows a resolved additional step when the under-travel starts (combined $+4.9 \pm 4.2$ mA). All quoted errors here are one standard error; the method is in Appendix A.

For orientation-dependent load the comparison is at matched IMU elevation. In the 90–150° IMU-angle window, the additional late-leg current contrast is $+15 \pm 25$ mA: no increase is detected. A window rise already exists on healthy up-legs (71 ± 10 mA in drive angle). Using drive angle instead gives an additional late rise of $+68 \pm 25$ mA, weak evidence; a direct paired test does not establish that the two coordinate comparisons differ. Matching elevation alone does not control azimuth or time. The *load thermistor* rises from 33.14 to 34.92°C on up-legs 1–51, while time and azimuth advance together, so this raster cannot separate the three as onset markers. It is not a measurement of motor or driver temperature. No battery voltage was recorded: approximately steady current alone cannot exclude supply-voltage sag. Whether the under-travel was an early stage of the later freeze remains open.

The failure. The last successful downward sweep completes at 01:41:39.38 UTC on 07-18. The next leg's step counts begin advancing at 01:41:40.45, 1.07 s later, with no comparable drive-current draw and the IMU elevation fixed (Fig. 5b). Across the 6 frozen legs in the two selected post-re-raise blocks, the mean extra draw is 1.1 ± 0.5 mA, or 0.34% of that morning's 0.313 A continuous running draw (0.7% at mean plus two standard errors). Thus there is no measurable elevation-drive current at the normal running scale, although the counts advance. This differs from the energised drive during the under-travel; it does not identify the failed component or establish a causal link between the events.

The antenna stayed near −176.05° (183.9°), facing the transmitter, to the end of the data at 03:08:49. The stillness criterion used for this memo's Fig. 5b first fires at 01:41:38 UTC; it is a detector boundary, not the physical failure instant. The current-and-command comparison above brackets the loss of response between sweeps. The field notes give 01:55, and legacy MEMO-002's trailing detector gave 01:42:32.

7 The elevation axle's tilt, from the box's IMU

Eq. (1) takes the elevation axle to be horizontal. It is not, and its tilt changes between raises. With a tilt τ of the axle out of the horizontal, the mount is

$$v_{\mathrm{ENU}} = R_z(\psi)\, R_y(\tau)\, R_x(\mathrm{el})\, R_z(\mathrm{az})\, v_{\mathrm{body}}, \tag{2}$$

where positive τ lowers the ψ end of the axle (bearing 307.84°; memo 005, `eigsep_data.beam_mapping.tx_fit`). The beam fit (`v0017`) fits one τ for the 07-17 raster, +4.74° (memo 005), and every fit that applies the beam elsewhere has used that value. Box-air's elevation IMU measures τ in every era, with three limits (§7.2): on a parked box the axle's direction is carried over from the raster, the sign is taken from the raster's beam fit, and an accelerometer bias along the axle is not measured. The numbers and figures of this section come from Christian Hellum Bye's analyses of the IMU readings (the IMU tilt analysis below) and of the IMU's elevation (the IMU elevation analysis) and from the night fit of memo 008, §5.1, which are not shared; their outputs are frozen, with SHA-256, in `inputs/tilt/` (Appendix A). The IMU tilt analysis works on `raw_binned_v0006`'s bin means of the `imu_el` accelerometer, with the firmware calibration's accelerometer bias removed from every reading, as the firmware removes it before it computes `el_deg` (below). Pointing table v6 carries the tilt per sample. This memo is on pointing table v5; v6 is built (`v6+4af5ce9`), not yet released, and described in §7.1.

The geometry. The IMU (a BNO08x-family sensor, like `imu_az`'s BNO085 of §5.1) is on a circuit board in an enclosure screwed to the top of the box. The elevation axle runs through the box, with the elevation motor inside it, so the box and the sensor turn together about the axle, and the sensor reads the direction of gravity in its own frame. As the box turns through a revolution, that direction sweeps a cone about the axle. If the axle is horizontal the cone opens into a flat circle in the plane perpendicular to the axle; if it is tilted by τ, every reading lies at τ out of that plane, on the same side. Two things follow from one elevation sweep. The axis of the cone, the normal $\hat{n}$ of the plane fitted to the unit readings $\hat{u}$, is the axle's direction in the sensor's frame. Each reading's angle out of the plane,

$$\tau = s \arcsin(\hat{u} \cdot \hat{n}), \tag{3}$$

is the axle's tilt at that moment, with a sign s fixed below (§7.2). The elevation is the reading's angle *around* the cone. That is what the firmware's `imu_el.el_deg` reports and the table's `el_deg` copies (§3); the component along the axle, which carries the tilt, has been discarded until now. On a parked box, with no sweep, $\hat{n}$ is carried from the raster (§7.2).

The sensor is turned 3.85° in the box. The raster's plane normal lies 3.85° from the sensor's x axis (Fig. 7c): the sensor is turned in the box by a constant angle. Read along the sensor's x, the tilt then picks up a component of the in-plane gravity that swings with elevation (scatter 2.85° over the raster; Fig. 7a, grey); read about $\hat{n}$ it is flat at +5.92°, with a scatter of 0.87° (red), or 0.58° outside a seven-minute dip (below). Three models of the raster's unit readings, with the accelerometer's bias across the axle removed as the firmware removes it (below; Fig. 7d):

- a constant tilt read by an aligned sensor (2 parameters: the tilt and an elevation zero): rms 2.86°;
- a constant tilt read by a sensor turned by a fixed rotation (4): rms 0.87°;
- the same freedom spent on a tilt that varies with elevation, $\tau_0 + A\cos\mathrm{el} + B\sin\mathrm{el}$, read by an aligned sensor (4): rms 0.91°.

On the re-raise, held out with the raster's parameters fixed (Fig. 7e), they score 2.05°, 0.25° and 0.27°. The held-out bins stop at 03:01 UTC: from 03:02 the box was coming down at the end of the deployment, and the 133 later bins, most of them above el = +100°, read −1.1° on average.

Figure 7(d, e) splits each residual into its component along the axle, $\Delta\tau$ (the tilt error), and around it, Δel (the elevation error). The two four-parameter models predict the same tilt to second order in the turn ($1 - \cos 3.85° = 0.2\%$): a turned sensor reads like a tilt that varies as cos el and sin el, with the turn's size as its amplitude, so $\Delta\tau$ cannot choose between them. Around the axle they differ at first order in the turn times the tilt, about 0.4° at the raster's tilt. That comparison needs the accelerometer's own bias removed first. The firmware's `imu_el` calibration (below) fits a constant bias in the plane of the readings, $(y, z) = (+1.89, -2.75) \times 10^{-3}\, g$, and subtracts it before it computes `el_deg`; left in, it adds an error around the axle that goes once round per turn (the bias itself is tested on the raster below, Fig. 8). On the raw readings (rms 2.86°, 0.89° and 0.93°), Δel is 0.17° rms for the turned sensor and 0.31° for the aligned cos / sin model, and both follow elevation. With the calibration's bias removed, and no parameter added, Δel is 0.04° for the turned sensor, with every 10° bin within 0.06° of zero, inside the noise; the cos / sin model keeps 0.28°, with bins up to 0.41°. On the held-out re-raise they are 0.008° and 0.10° (raw: 0.11° and 0.19°). The tilt does not move: +5.92° (raw: +5.92°). A measurement made independently also favours the turned sensor. The firmware's `imu_el` calibration was fitted in the lab on 07-09, with the box hung differently, to a 12-stop elevation sweep, and is stored in the header of 1,485 of the 2,427 phase-C files. It finds the sensor turned by 3.72°: a property of the mount, not of the hang. A box whose tilt followed elevation with exactly the sensor's misalignment would be a coincidence. The turn rotates the sensor, not the box: it does not tilt the antenna, and the firmware's `el_deg`, computed after the calibration's rotation, already removes it.

The fitted models' residual is not sensor noise (0.083°). It lies along the axle: $\Delta\tau$ is 0.87° rms for the turned sensor, against 0.04° in Δel, and 0.60° without the raster's dip (below). What remains is the tilt changing in time, the dip and the wander of §7.2, not a model error that follows elevation; the bin means of $\Delta\tau$ stay near zero (Fig. 7d).

The calibration's bias, tested on the raster. An elevation sweep can measure two of the bias's three components, those across the axle; the calibration pins the third to zero (§7.2). That `el_deg` is reproduced once the calibration's bias is removed shows that the same bias is used here, not that it is right. The raster tests

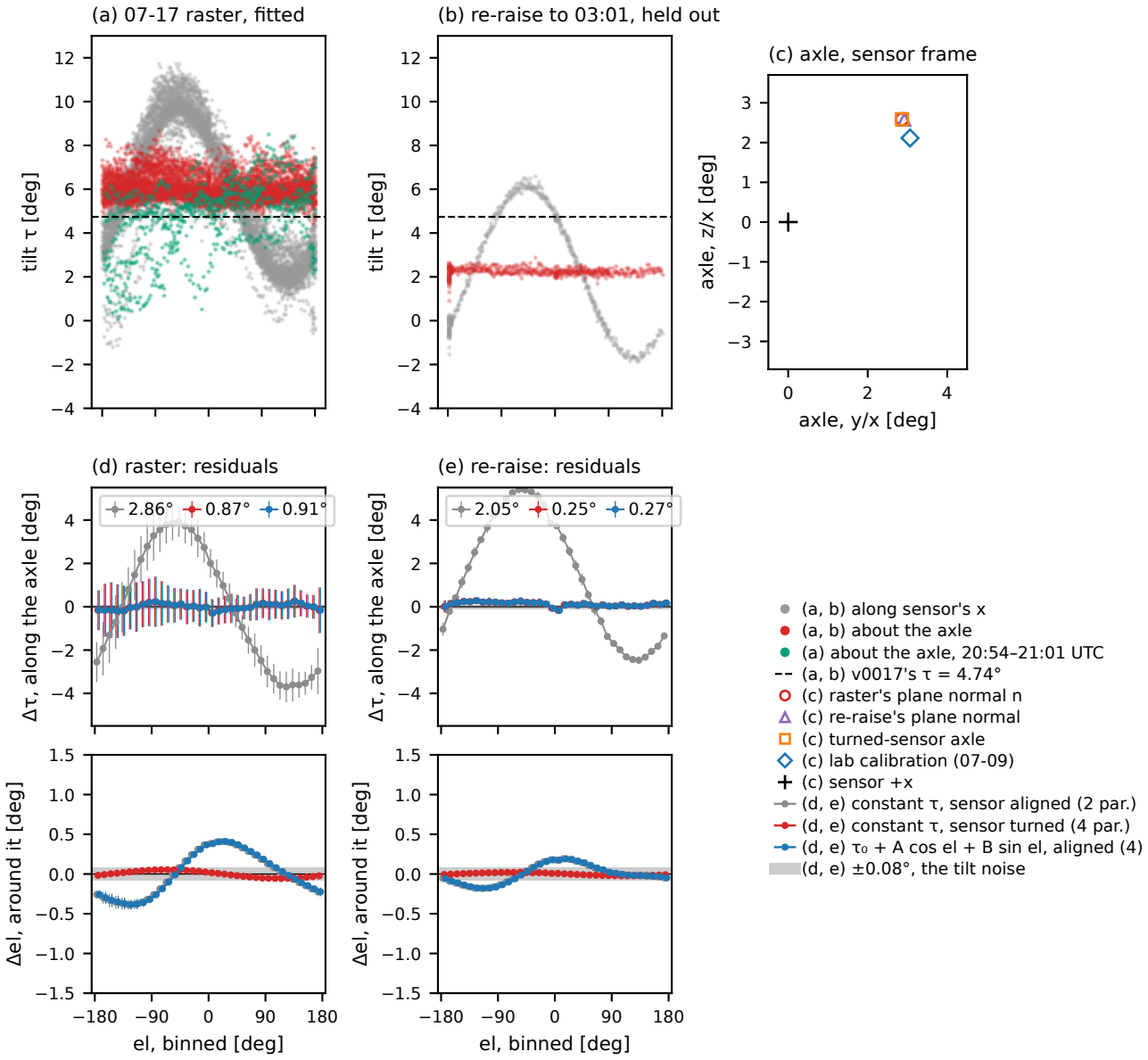


Figure 7: The axle from the IMU's readings (the IMU tilt analysis's bin means of `raw_binned_v0006`; quasi-static bins, $\left|\,|a|/g - 1\right| \leq 2\%$, with the binned elevation's wrap glitches removed; drawn by this memo's `tilt_axis.py`). (a) The tilt of each of the 6,517 bins of the 07-17 raster against the binned elevation, read along the sensor's own x axis (grey), as if the sensor were aligned with the axle, and about the axle fitted as the plane normal $\hat{n}$ (red, Eq. 3); green: the same for the bins in 20:54–21:01 UTC; dashed: the beam fit's τ. (b) The same for the 574 bins of the 07-18 `~91m-reraise` era before 03:01 UTC, with $\hat{n}$ from the raster. (c) The axle's direction in the sensor frame, as its angles from the sensor's $+x$ toward $+y$ and $+z$: the raster's and the re-raise's plane normals, the turned-sensor model's axle, and the axle of the firmware's `imu_el` calibration, made in the lab with the box hung differently (07-09 19:41 UTC). (d, e) The residuals of the three models, fitted to the readings with the calibration's accelerometer bias removed, in 10° bins of elevation, along the axle ($\Delta\tau$, the tilt error) and around it (Δel, the elevation error): each bin's weighted mean, with its weighted spread as an error bar (the models offset by 2° in el); grey band: ±0.083°, the tilt noise between neighbouring samples. (d) The raster (fitted); (e) the re-raise (held out: the raster's sensor parameters fixed, only the tilt refitted). The two aligned models' Δel coincide (grey under blue). Legends: each model's rms angle.

it (Fig. 8). At rest the sensor reads $|a - b| = g$ at every elevation if b is its bias; an error in b across the axle leaves a term in $|a - b|$ that goes once round per turn, with the error's size as its amplitude. Over the raster's 6,517 bins, a full turn, that amplitude is 1.99 ± 0.07 with no bias removed and 1.62 ± 0.07 with the calibration's, in units of $10^{-3}\,g$ (errors: a bootstrap over 50 time blocks of 72 s). The calibration's own fit, a circle through the readings, made on the raster gives $(y, z) = (+0.47 \pm 0.07,\ -1.94 \pm 0.07) \times 10^{-3}\,g$ against the calibration's $(+1.89, -2.75)$: the same signs, smaller in both components. Held out, a bias fitted on one half of the raster in time leaves 0.55 on the first half and 0.55 on the second, against 1.50 and 1.77 with the calibration's and 1.93 and 2.09 with none (errors 0.07–0.12).

The readings' magnitude also varies twice per turn, by $(9.6 \pm 0.1) \times 10^{-3}\,g$ (Fig. 8a–c), 6 times the once-per-turn term the calibration's bias leaves. A bias of this size makes no such term. A difference of scale between the sensor's two axes across the axle, or an angle between them other than 90°, would, and the calibration's model, one bias and one scale, has neither. The fits above carry the term, so that it does not pull the bias where the readings cover the turn unevenly: without it the firmware's fit gives $(+0.63 \pm 0.15,\ -1.82 \pm 0.56) \times 10^{-3}\,g$ on the raster.

The IMU elevation analysis fits both to the raster's readings. The two axes are 91.08° apart ($\psi = -1.078 \pm 0.007°$ from orthogonal), which carries 95 % of the term's power; a difference of scale, -4.3×10^{-3}, carries the rest. Fitted on one half of the raster in time, this model removes the term on the other: from 9.56 to 0.08 ± 0.08 on the first half and from 9.59 to 0.08 ± 0.09 on the second, in $10^{-3}\,g$. That tests the parameters' stability, not the model, which a free twice-per-turn term in the magnitude would match as well. At rest, the 12 holds at a fixed elevation whose reference, the box parked level, was taken within 20 min show the term, $(9.6 \pm 1.2) \times 10^{-3}\,g$ with its peak at el = +36.7° (on the raster +38.4°), but that selection was made after seeing the data. The full set of 21 holds, the test stated beforehand, does not settle it: $(6.4 \pm 3.7) \times 10^{-3}\,g$, with a poor fit ($\chi^2 = 67.6$ for 17 degrees of freedom).

If the term is the sensor's, the elevation it reads is off by $0.553 \pm 0.004°$ twice per turn. The table's `el_deg`, in v5 as in v6, is the accelerometer's elevation and carries this in full; at the 07-17 night's hold it is +0.54° relative to its average over the raster's turn. The motor is an independent check. Over the raster's 44 legs, with one offset per leg, the IMU's elevation less the motor's has a twice-per-turn term whose cosine part, $0.66 \pm 0.04°$ against 0.55° predicted, agrees to 21 %, but whose sine part, $0.60 \pm 0.06°$ against −0.08°, carries 0.68° that the model does not explain (0.64–0.70° on rising and on falling legs and on each half of the raster). The drive is a stepper on a worm gear, which would not be expected to make a twice-per-turn error, so the IMU's error may be up to 0.9°. This release does not correct it: the error is carried as a systematic of the elevation, and a corrected table waits for the calibration's 12 stops at rest, which the files do not keep. An elevation error of 0.55° changes beam v0018 by 1.2–2.3 % of its peak (the largest change over its −10 dB region, 64–240 MHz), below the beam fit's held-out fractional residual at every frequency it scores (at least 3.9 %).

The raster's estimate is not a new calibration. Its bins were taken with the box turning, at a median $5.2° \text{ s}^{-1}$ (bins 0.54 s apart, 99 % of them single integrations), and motion that follows elevation enters a circle fit as a bias. Steady rotation adds a centripetal acceleration $\omega^2 r$ toward the axle, constant in the sensor frame, which mimics a bias and has the same sign on legs turning up and down. With the IMU less than 0.5 m from the axle it is at most $0.43 \times 10^{-3}\,g$ at the median rate, 26 % of the discrepancy with the calibration's bias (below). Because ω^2 varies with elevation by at most 9 % once or twice per turn, its part twice per turn is at most $0.037 \times 10^{-3}\,g$: rotation cannot make the twice-per-turn term. Legs turning up and down, fitted apart, agree to $0.16 \times 10^{-3}\,g$, which excludes only motion that reverses with the turn. The estimate also changes within the hour, from $(+0.71, -1.81)$ on the first half to $(+0.23, -2.09)$ on the second, and the sensor's bias can drift with temperature between the calibration on 07-09 and the raster on 07-17. What the raster shows is a discrepancy of $1.62 \times 10^{-3}\,g$ with the calibration's bias. The firmware's bias is kept: it defines the table's elevation, and the readings of this section are on it. Replacing it with the raster's would change the elevation by a once-per-turn term of 0.09°; replacing it with the raster's or with none moves the night's mean tilt by at most 0.014° and an era's by at most 0.016° on the same bins, against the night mean's error of 0.13°.

A dip in the raster. From 20:54 to 21:01 UTC (749 bins, 11% of the raster's) the tilt averages +4.78°, with a scatter of 1.59° and a lowest bin at +0.07°, against +6.07° in the rest of the raster (Fig. 7a, green). The bins are quasi-static, $\big||a|/g - 1\big| \le 2\%$, but that cut is second order in a horizontal acceleration and admits ones that would read as a tilt of degrees; Fig. 9 counts the dip among the stretches where the box was being moved. The dip coincides with the end of the fastest divergence between the potentiometer and the motor, 20:41–20:54 (§4), which this memo attributes to the potentiometer, not the platform. Nor is it the elevation under-travel

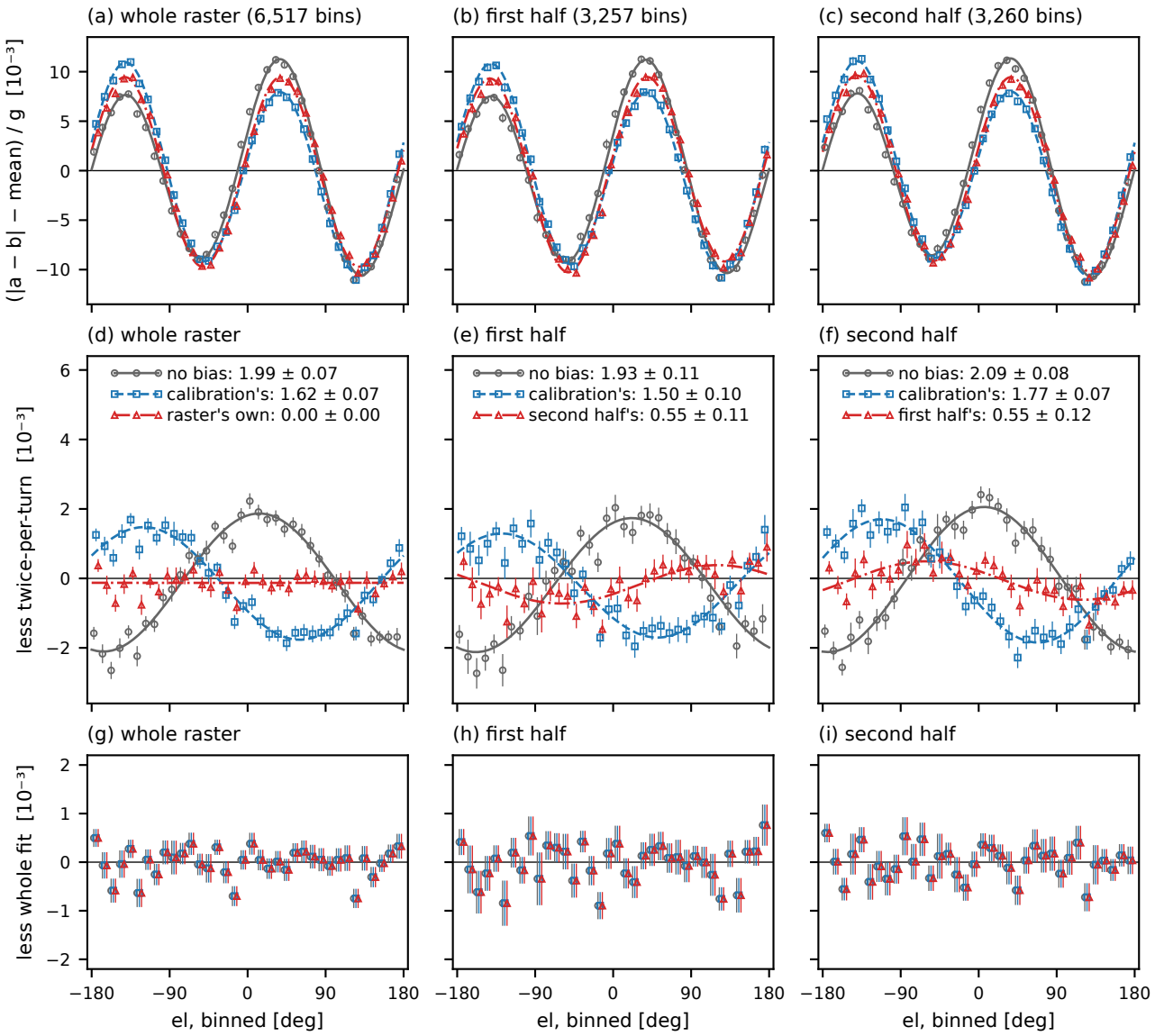


Figure 8: The calibration's accelerometer bias tested on the 07-17 raster: the raw readings of its 6,517 kept bins (the bin means of Fig. 7a, a full turn in elevation; drawn by this memo's `tilt_axis.py`). (a–c) The readings' magnitude with a candidate bias b removed, $(|a - b| - \text{its mean})/g$ in units of 10^{-3}, in 10° bins of elevation: each bin's mean, weighted by rows per bin, with the standard deviation over a block bootstrap as its error bar (2,000 resamples of time blocks of 72 s; the candidates offset by 2.5° in el). Grey circles and solid curve: no bias removed; blue squares, dashed: the calibration's bias; red triangles, dash-dotted: the bias fitted to the raw readings (left: on the whole raster, so its once-per-turn term is zero by construction; centre and right: fitted on the other half, a held-out test). Curves: each candidate's fit of a constant, a once-per-turn and a twice-per-turn term. (d–f) The same less the fitted twice-per-turn term, with the fitted once-per-turn term; legends: its amplitude and bootstrap error, in $10^{-3}\,g$. (g–i) The same less the whole fit. Left column: the whole raster; centre and right: the raster's first (3,257 bins) and second (3,260) half in time. The twice-per-turn term dominates (a–c) and is the same for every candidate. In (d–f) the calibration's bias leaves a once-per-turn term nearly as large as no bias does, and a bias fitted on the other half of the raster leaves a smaller one. The scatter of single bins about the whole fit is $3.2\times10^{-3}\,g$ rms.

of §6: the dip falls on legs 23–29, among the early legs whose residual leg medians stay within ±4.1°, and the under-travel starts at leg 45, at 21:19 UTC, 18 min after the dip ends. What turned the box about the axle is not known. Beam product v0018 (memo 005, §5), fitted with τ held at the IMU's raster mean, predicts the held-out samples in the window as well as its fitted-tilt variant does: a per-tooth fractional rms of 0.0627 against 0.0626 (mean over teeth); outside it, 0.0770 against 0.0763. Both, and HFSS (0.0798 against 0.0959), score better in the window than outside, so that difference is in the directions sampled, not the tilt. Over the beam fit's kept samples the IMU reads +4.77° in the window and +6.07° outside; refitting the tilt alone with the HFSS beam gives +4.88° without the window's samples and +4.05° on its 501 training samples alone, with no error computed: consistent with a radio tilt that follows the dip by less than the IMU's.

The tilt in each height era. Read about $\hat{n}$, the tilt changes at each raise and then settles (Fig. 9, Table 5). Table 5 gives three means per era. The first two are over each source's own samples in the era: pointing table v6's quasi-static rows, +5.34° in `~91m` (the 07-17 raster's era), +0.32° in `~87.5m` (the 07-17 night's), +2.09° in `~91m-reraise` and −0.58° in `~30m`, and the IMU tilt analysis's bins. On the same bins, v6's rows inside its bins, the two agree to 0.022° per era. Their whole-era means differ by more, because v6 has rows where the analysis has no bin: in `~2m`, its bins start at 07-15 00:31 UTC and hold 54,777 of v6's quasi-static rows, and v6's others include phase A on 07-13, with the 290 rows of undefined elevation (§7.1). These whole-era means include the box settling after a raise: after the raise to `~91m` the parked box read +3.9° in the era's first 15 min and +5.5° from an hour later until the raster. The third column leaves out each era's first hour and the stretches where the box was being moved: +5.79°, +0.15°, +2.01° and −0.27° in the same four eras. Over the raster window itself the tilt is +5.92°, the number this section compares with the beam fit; over the samples the beam fit's geometry stage used, it lies +1.19±0.13° (a block jackknife) above the fit's +4.74°. Over `fit_v0025`'s 362 night bins (memo 008; 04:12–11:24 UTC) it is +0.18°, with a scatter of 0.54° and a jackknife error of the mean of 0.13°, an error that holds if the tilt was constant through the night. It was not: over the night's 15.4-h hold (03:22–18:48 UTC) the tilt's 1-h running mean wanders by 0.56° rms. If the frame turned, the night's mean is known to 0.13° but a constant τ misdescribes the night; if the accelerometer's bias drifted, the mean is off by an amount not known. We quote $+0.18 \pm 0.6°$: the wander over the whole hold, taken as an allowance for the night fit's mean, with the axis's 0.15° in quadrature. A constant accelerometer bias along the axle, 0.06° per 0.01 m s^{-2}, would move every bin alike and is not in the number; one the size of the firmware calibration's biases across the axle would be 0.11–0.16° (§7.2). The night fit (memo 008, §5.1) held τ at +0.19°, the night's mean from the readings before the bias was removed. With the bias removed the mean is +0.18°, 0.015° lower, against the mean's error of 0.13°, the quoted ±0.6° and the ±0.5° steps of the fit's tilt band. The fit moves to the corrected value at its next rerun; its scan, its χ^2 and its residuals below are as it ran. It drops the 44 bins in which the IMU shows the box moving in the wind; over the 318 bins it keeps, the tilt is +0.18°. The night's bins range from −1.1° to +1.5°, and the tilt keeps wandering through the rest of the hold (Fig. 9a). The box hung about +5.92° out of level along the axle during the raster and within about 0.6° of level on the night.

Table 5: The axle tilt per height era: pointing table v6's quasi-static rows (mean and scatter) and the IMU tilt analysis's bins (mean weighted by rows per bin), each over its own samples in the era, which are not the same (on the same bins they agree to 0.022°; text); and the analysis's bins from a settled start, leaving out each era's first hour and the stretches where the box was being moved (Fig. 9). The `~91m` era starts at 18:49 UTC, before the raster (20:26:00).

height era	quasi-static rows	τ, v6 [deg]	scatter, v6 [deg]	IMU bins	τ, IMU bins [deg]	τ, settled [deg]
~`2m`	468,919	+2.56	2.67	1,497	+1.88	+1.81
~`30m`	119,474	−0.58	2.87	2,774	−0.55	−0.27
~`87.5m`	107,709	+0.32	1.51	1,097	+0.36	+0.15
~`91m`	17,567	+5.34	0.95	7,578	+5.37	+5.79
~`91m-reraise`	12,450	+2.09	0.70	707	+2.09	+2.01

The radio data. Two radio fits test the tilt. First, the beam fit on the raster (memo 005), with τ free as a geometry correction, prefers +4.74°, below the IMU's raster value; the sign of Eq. (3) is chosen to match it (§7.2), so this is no test of the sign. Second, the night. A two-column sky model is fitted, per frequency, to the 318 bins of memo 008's night fit (`fit_v0025`'s 362 less the 44 in which the IMU shows the box moving in the

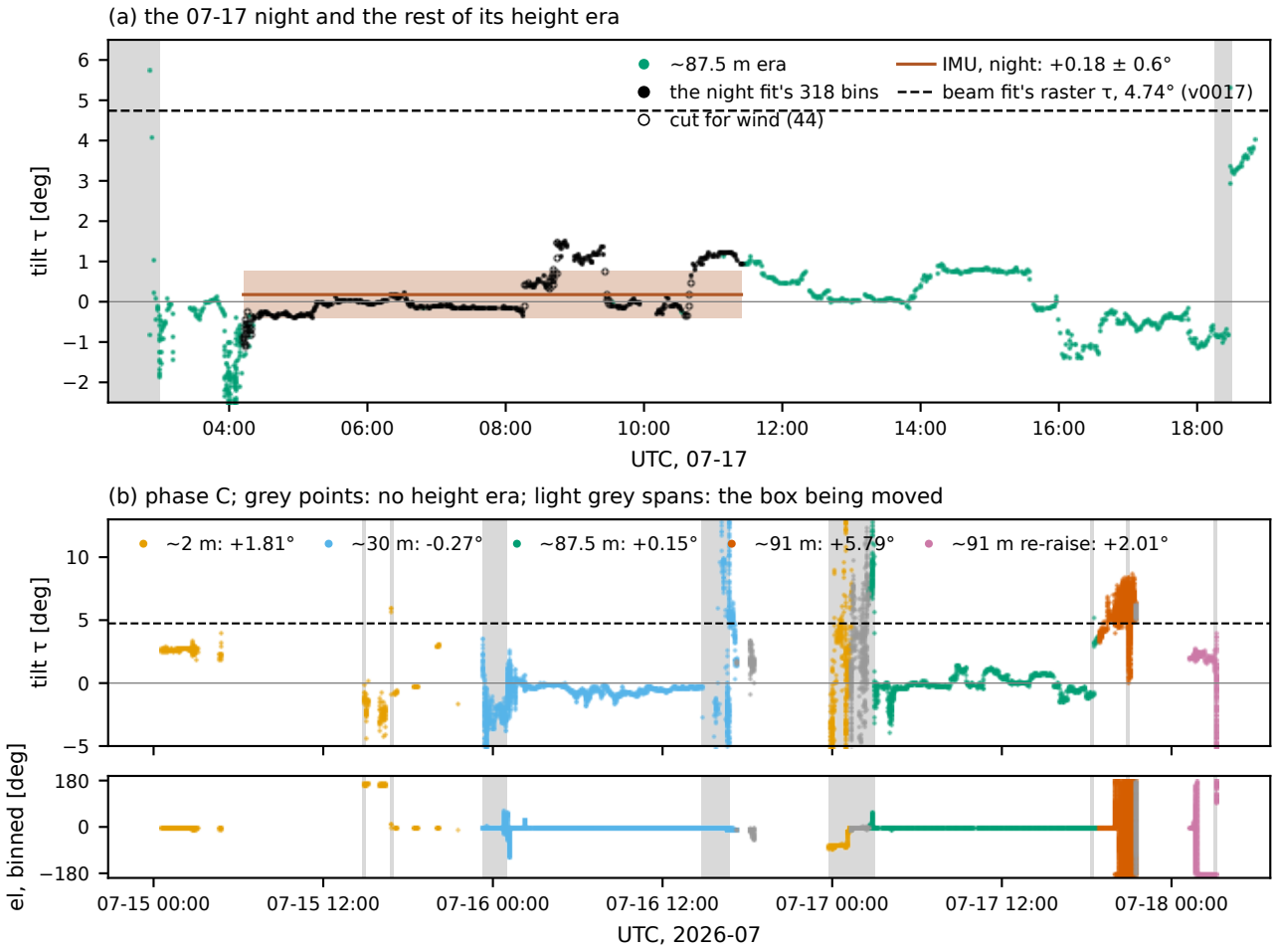


Figure 9: On the 07-17 night the box was level to within about 0.6° (IMU), 4.6° from the raster tilt the beam fit applied to it. (a) The tilt about the axle (Eq. 3) of the `~87.5m` era's quasi-static bins (the IMU tilt analysis's bin means of `raw_binned_v0006`; drawn by this memo's `tilt_eras.py`): `fit_v0025`'s 362 night bins in black, filled for the 318 the night fit keeps and open for the 44 its wind cut removes (memo 008, §5.1), the rest of the era in green; brown line and band: the IMU's night value, $+0.18 \pm 0.6°$; dashed: the beam fit's $+4.74°$, fitted on the raster. (b) Phase C, coloured by the mode table's height era (grey: bins the mode table gives no height era); legend: each era's mean from a settled start (Table 5). Bottom: the binned elevation, the instrument's state. Light grey spans: the box being moved, the 8 stretches of 15-min windows whose tilt spread (5–95%) exceeds 3° (raises, a lowering, handling on the ground, the raster's dip and the fall). Their mean $\left||a|/g - 1\right|$ is 0.1–3.0%, against 0.10% over the night, mostly within the 2% quasi-static cut: real pitching and swinging on the ropes and an apparent tilt from horizontal acceleration (second order in $|a|$) both enter, and the accelerometer cannot separate them.

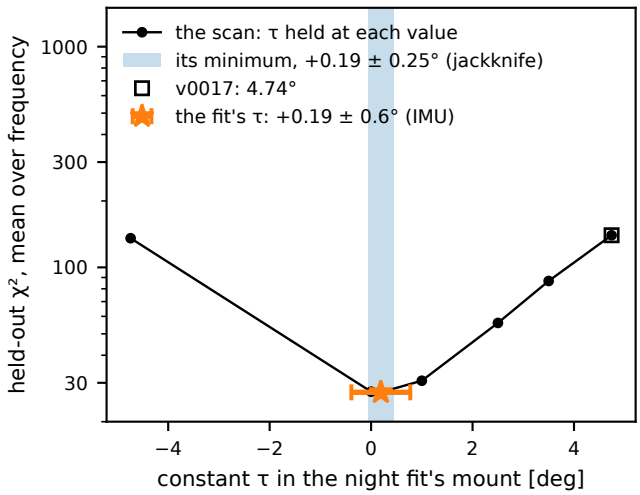


Figure 10: The 07-17 night's held-out χ^2 (the night fit's two-column plain fit on the 318 bins its wind cut keeps; mean over 43 frequencies) against a constant tilt in the night fit's mount: a scan, in which each τ's beam orientation is recomputed and θ and P refitted. Blue band: the scan's minimum and its block-jackknife error; star: the tilt the headline fit holds, +0.19°, the IMU's mean over `fit_v0025`'s night bins on the readings before the bias was removed (with it removed, +0.18°; text), with the IMU's quoted error; square: the beam fit's raster tilt. The floor, 27.2 at the fit's tilt, is far above 1: the model is not noise-limited (the residual is memo 008's subject), and the minimum's error is the jackknife's, not a $\Delta\chi^2 = 1$ interval.

wind) at its 43 frequencies: the night fit's plain fit, $y = \theta A_\tau + P$, with θ a dimensionless scale of the sky, P an additive offset in K, and $A_\tau = S_\tau + T_{\rm CMB}(1 - f_\tau)$, where S_τ is the GSM2016 sky without the CMB through the native HFSS beam on the mount of Eq. 2, above the terrain horizon, f_τ the fraction of the beam below that horizon and $T_{\rm CMB} = 2.725\,$K. It is trained on alternate 30-min blocks and scored on the others. Unlike `fit_v0025`, it has no free sky field that could absorb a mount error. The model's τ is scanned (Fig. 10): at each constant value the beam's orientation is recomputed, and θ and P are refitted on the training blocks and scored on the others. The held-out χ^2, averaged over frequency, is 139.8 at the beam fit's +4.74°, 27.4 at $\tau = 0$ and 27.2 at the fit's +0.19°, the IMU's value as the fit took it (above). Its minimum is at $\tau = +0.19 \pm 0.25°$ (the vertex of the parabola through the scan's lowest point and its two neighbours; a delete-one-block jackknife over its 15 half-hour blocks). The raster's tilt, carried to the night, predicts the held-out data 5.1 times worse than the IMU's. That is what the night shows: the raster's tilt does not carry over to it. The scan is consistent with the IMU's value and with level (27.2 against 27.4), and does not separate them. Its minimum depends on the wind cut, which uses the IMU: on all 362 bins it is $0.00 \pm 0.41°$ (the same jackknife). The headline night fit (memo 008) fixes τ at the IMU's value, not at the scan's minimum: the IMU is independent of the sky data, and a τ taken from the scan would absorb other time-dependent errors of the model. θ at $\tau = 0$ differs from θ at the IMU's value by -0.28% (weighted mean over frequency; -0.57 to $+0.15\%$; the night fit).

Data, model and residual. The model is the headline night fit (memo 008, §5.1): the two-column fit with memo 004's load term, on the 318 bins its wind cut keeps and 43 frequencies, at the fit's $\tau = +0.19°$ (the IMU's, above) and, as a variant, at the beam fit's +4.74°. Only the mount's tilt differs between the two. Their waterfalls of data, model and residual, time down the page and frequency across, are in memo 008, Fig. 4 (and at 70.31, 148.44 and 199.22 MHz in its §5.1); this memo refits them from the night fit's products (Appendix A) and quotes the residual over σ, the in-bin scatter with a multiplicative white term (memo 008's $\sigma_{\rm tot}$). With the beam fit's tilt the residual is a smooth drift across the night at every frequency, positive early and negative late, consistent with a mistimed Galactic transit. Its rms is $12.1\,\sigma$ averaged over frequency (worst $18.9\,\sigma$), with a lag-one autocorrelation between consecutive bins of 0.98. With the IMU's tilt the model follows the transit: rms $4.6\,\sigma$ (worst $12.4\,\sigma$), lag-one autocorrelation 0.85. $\tau = 0$ does as well on held-out data (27.4 against 27.2). What remains is not the tilt's: it is the same with the IMU's per-bin tilt (held-out χ^2 27.3), so the tilt's wander within the night does not explain it. It lies mostly above 200 MHz (memo 008, Fig. 4, the IMU's residual). At 70.31 MHz it is a scatter of rms $4.3\,\sigma$ (up to $13\,\sigma$) that is not white (lag-one 0.80); at 199.22 MHz a slow

wave (lag-one 0.90), whose 30-min running mean runs from $+7\,\sigma$ at the night's start (04:20) to $-6\,\sigma$ near 09:54. Memo 008, §5.1, takes them up.

The raster's fitted tilt. The antenna is bolted to the mount on the box, so the axle tilt the beam sees is the IMU's. Beam product v0018 (memo 005, §5) holds τ at the IMU's raster mean; its fitted-tilt variant, refitted on pointing table v6 with τ free, gives $+4.74\pm0.23°$ (a delete-one-stripe jackknife over its training stripes). The two fits differ in more than τ: the variant fits τ alone and holds the azimuth, elevation and polarization corrections at zero (`axle_tilt`), and v0018 fits those three with τ held at the IMU's $+5.91°$ (`polarization_az_el`). Held out, the variant predicts the validation stripes no better than v0018: 0.0724 against 0.0722 (mean over 64 teeth of the per-tooth fractional rms), worse on 37 teeth. Only the geometry stage, which uses the HFSS beam, prefers the fitted τ (the median over teeth of the validation fractional rms, 0.0758 against 0.0792); the empirical beam fitted at the IMU's τ absorbs the difference, a change of 0.074 dB rms on the sampled directions and at most 0.28 dB within 20° of the boresight. So the raster cannot separate a change of tilt from a small change of the beam's shape, and its fitted tilt says nothing about the mount. The fitted-tilt variant is kept as the documented alternative for users (memo 005, §5.6).

7.1 Pointing table v6

Pointing table v6 (`v6+4af5ce9`, built at `data-analysis 4af5ce9`, whose builder `main` carries unchanged at `912ae15`; 1,227,956 rows in the same order as v5) adds:

- `tilt_deg`, the tilt of Eq. (3) per sample, from the integration's averaged `imu_el` accelerations through the firmware calibration's bias, with $\hat{n}$ fitted on the raster;
- `tilt_sigma_deg` = 0.171° on every row: the per-sample noise, 0.083° from static neighbours, and 0.15° for the axis (§7.2), in quadrature. It leaves out the tilt's wander during a hold and any accelerometer bias along the axle, so it is not the error of a tilt averaged over a hold (the night's is ±0.6°, §7);
- the flag `IMU_NOT_QUASI_STATIC` (16384), where $\big||a|/g - 1\big| > 2\%$ (7,707 rows): the box is swinging, and neither the tilt nor the elevation is the axle's. It makes a row suspect;
- the flag `EL_UNDEFINED_TILT` (65536), where $|\tau| > 60°$ (290 rows): gravity lies close to the axle, so the elevation, the reading's angle around it, is undefined. All are on 07-13 in the `~2m` era, most likely the box being handled on the ground. It makes a row suspect.

It also fixes an elevation wrap. The recorder averages each floating-point metadata field linearly over an integration's samples (`eigsep_observing/io.py`, `_avg_sensor_values`). Near el = ±180°, readings of +179.9 and −179.9 average to about 0, or ±60 with three samples, and the table copied that value. The averaged accelerations are vectors and are unaffected, so v6 computes `el_deg` from them through the calibration, as $\mathrm{atan2}(g_y, g_z)$. That corrects 314 rows (242 in the `~91m` era, 57 in `~2m`, 13 in the re-raise and 2 in no height era), flagged `EL_WRAP_CORRECTED` (32768). `EL_SOLUTION_GLITCH` stays set on them, since it is true of the firmware's reading, but no longer makes a corrected row suspect; the other criteria decide. Of the 305 rows that had wrapped by 10° or more, 275 are now `ok`; the 9 corrected by 1–10° stay suspect. The files hold only the per-integration means, so the individual samples cannot be recovered.

Uncommanded motion (`UNCOMMANDED_MOTION`, 32) is now detected on the elevation from the averaged accelerations rather than on the firmware's, whose wrap glitches had triggered it: 2,750 of v5's 9,560 flagged rows lose it, and 0 rows gain it. Over the table 7,185 rows go from `ok` to suspect and 2,449 from suspect to `ok`; `ok` rows number 582,743 (v5: 587,479).

In the raster window (6,990 rows, among them the 60 solver-jump rows that §6's 6,930 leave out), the 60 wrap rows lie near el = ±180°, where the antenna faces the transmitter (183.7°), and 60 of them are now `ok`. The beam fit keeps only `ok` rows (`fit_beam.py`), so `v0017`, on v5, had none of them. 31 rows that v5 had `ok` are now suspect, 31 of them as not quasi-static. The raster's `ok` rows go from 6,788 to 6,817.

Unchanged: `az_deg` and every column other than `el_deg`, `flags` and `quality`. On the 271,328 static rows, where v5's elevation moves less than 0.02° from both neighbours in the same file, the new `el_deg` equals v5's to 3×10^{-5}° rms (worst 0.005°, 0 rows beyond 0.02°). On the 191,396 rows moving 0.02–0.1° per row, the mean of the samples' angles (v5) and the angle of their mean vector (v6) can differ: worst 0.039°, 5 rows beyond 0.02°. Across the table the elevation moves by more than 0.02° on 1,175 rows, the wrap rows among them.

7.2 Caveats

- **The sign is taken from the raster beam fit.** The calibration cannot fix it: the sign of Eq. (3) depends on whether the BNO08x reports specific force ($+g$ up at rest) or the gravity vector, and the firmware's own fit is the same either way. The software emulator of the IMU in `picohost`, which tests the host code, assumes specific force; the sensor's convention is not confirmed. The beam fit, which uses no IMU, prefers a positive tilt on the raster, and s is chosen to match it. The night does not constrain the sign: its tilt, $+0.18 \pm 0.6°$, is within its allowance of zero.
- **The axis is a 0.15° systematic.** Two independent field sweeps give the plane normal: the raster's lies 3.85° from the sensor's x, and the re-raise's, before the fall (574 bins over a full revolution), 3.89°; they are 0.06° apart (Fig. 7c). Leaving the raster's dip out moves its normal by 0.14°. In quadrature that is 0.15°, which changes a tilt by at most as much; on the night the two normals give the same tilt (+0.174° and +0.184°, on the 374 quasi-static `~87.5m` bins in 04:12–11:24 UTC). The firmware's lab calibration (07-09, the box hung differently) lies 0.51° from the raster's normal and 0.48° from the re-raise's, and would put the night at +0.65°. Its axle comes from a fit to the motor's commanded angles that assumes a horizontal axle, so the lab hang's tilt and the motor's errors enter it; it is not used for the axis. The night's radio minimum (+0.19°) is also closer to the plane normal's value. v6 uses the raster's plane normal.
- **An accelerometer bias along the axle** enters the tilt directly, about 0.06° per 0.01 $\mathrm{m\,s^{-2}}$. An elevation sweep turns the sensor about the axle and so never moves that component: the bias along the axle cannot be measured from one. The firmware's calibration fits 0.019 and 0.027 $\mathrm{m\,s^{-2}}$ in the two axes across the axle; a bias of that size along it would be 0.11–0.16°. The two fitted components are tested on the raster (§7, Fig. 8): the raster's own fit differs from them by $1.62\times10^{-3}\,g$, which moves the night's tilt by at most 0.014°.
- **The tilt wanders during a hold.** Over the night's 15.4-h hold (03:22–18:48 UTC, the elevation drive parked; 906 binned minutes) the tilt scatters by 0.86° about its mean, and its 1-h running mean by 0.56°, far more than the per-sample noise (0.083°). Whether the frame turned slowly or the accelerometer's bias drifted is not known: the first leaves the night's mean known to 0.13° but a constant τ misdescribes the night, the second biases the mean by an amount not known. The wander over the whole hold is the night's quoted allowance.
- **The axis is carried from the raster to the parked eras.** A parked box gives no sweep, so $\hat{n}$ there is the raster's, which assumes the sensor did not move in its enclosure between eras. The re-raise's own sweep gives the same axis to 0.06°, and the re-raise held out with the raster's axis scores 0.25° rms; the night and the earlier eras have no such test.
- **The raster's fitted tilt says nothing about the mount.** The beam fit's +4.74° differs from the IMU's +5.92°, but refitted on v6 it predicts held-out data no better than the IMU's tilt: the raster cannot separate a change of tilt from a small change of the HFSS beam's shape. The antenna is bolted to the mount on the box, and beam v0018 holds the tilt at the IMU's (above; memo 005).

8 Antenna phase center lever arm, platform stability, and elevation flexure

The mount model (Eq. 1) maps vectors from an idealized antenna body frame to local ENU. In the physical assembly, the antenna electromagnetic phase center (the dipole feed point and balun) does not sit on the mechanical gimbal axle: it is displaced by an offset vector $\mathbf{r}_{\mathrm{pc}} = [x_{\mathrm{pc}}, y_{\mathrm{pc}}, z_{\mathrm{pc}}]^T$ in the antenna body frame (boresight $+z$, dipole arm $+x$). Because the transmitter sits near nadir at el $\approx 183.7°$, the unit line of sight from the antenna to the transmitter is nearly vertical, $\hat{\mathbf{k}}_{\mathrm{tx}} \approx [0.046, 0.045, 0.998]$ in ENU. Under the roll mount rotation $R(\mathrm{az}, \mathrm{el}, \psi) = R_z(\psi)R_x(\mathrm{el})R_z(\mathrm{az})$, the line-of-sight path length to the transmitter modulates as

$$\Delta d(\mathrm{el}, \mathrm{az}) = \hat{\mathbf{k}}_{\mathrm{tx}} \cdot R(\mathrm{az}, \mathrm{el}, \psi)\, \mathbf{r}_{\mathrm{pc}} \approx z_{\mathrm{pc}} \cos(\mathrm{el}) + [x_{\mathrm{pc}} \sin(\mathrm{az}) + y_{\mathrm{pc}} \cos(\mathrm{az})] \sin(\mathrm{el}) + d_0. \tag{4}$$

A complete elevation revolution sweeps cos(el) between +1 (zenith) and −1 (nadir), producing a peak-to-peak path modulation of $2z_{\mathrm{pc}}$. The transverse terms $x_{\mathrm{pc}}, y_{\mathrm{pc}}$ modulate this path with azimuth as the antenna rolls.

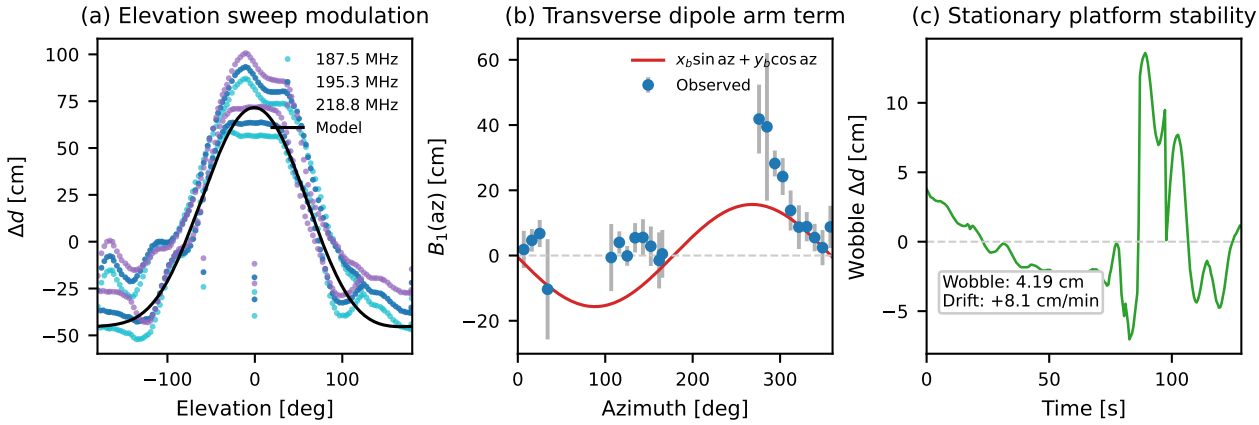


Figure 11: (a) Path length modulation Δd along the line of sight to the canyon transmitter across three comb frequencies (187.5 MHz, 195.3 MHz, and 218.8 MHz) during one elevation sweep of the 07-17 raster (`corr_20260717_203032Z.h5`, az $\approx 198°$), with the fitted 3-D mount model (black). (b) Fitted transverse lever arm term $B_1(\mathrm{az}) = x_b \sin \mathrm{az} + y_b \cos \mathrm{az}$ against commanded azimuth across the 22 clean raster sweeps (error bars: residual path rms), confirming the -15.7 cm offset along the dipole arm ($+x_b$) and near-zero (-0.5 cm) transverse offset ($+y_b$). (c) Platform stability during a 128 s stationary dwell (`corr_20260717_155704Z.h5`): linear thermal/cable drift (+8.1 cm/min) subtracted, leaving high-frequency wind wobble of 0.75 cm rms on < 5 s timescales (4.19 cm over the 128 s dwell, 9.8° phase rms at 195.3 MHz).

Multi-frequency geometric consistency. The cross-correlation between the stationary ground antenna (`box-gnd`) and the raised antenna (`box-air`) on the transmitter comb teeth (memo 001) yields an interferometric phase $\phi(\nu) = 2\pi\nu\,\Delta d/c + \phi_0$. Across 187.5–218.75 MHz (channels 768 to 896), the phase excursion per elevation sweep scales strictly with frequency: on reference file `corr_20260717_203032Z.h5`, the unwrapped phase span increases from 313.5° at 187.5 MHz to 377.7° at 218.75 MHz, yielding an invariant geometric path modulation of 139.3–143.8 cm peak-to-peak (Fig. 11a). This confirms that the modulation represents non-dispersive path geometry, not an electronic dispersion or instrumental phase artifact.

3-D lever arm measurement. Fitting Eq. (4) across the 22 clean raster sweeps outside the cross-polarization null (Table 1) decomposes the 3-D phase center position:

- **Boresight offset** z_{pc}: $+55.6 \pm 6.6$ cm along $+z$. When pointing at zenith, the dipole sits ≈55.9 cm above the elevation axle, furthest from the canyon floor; at nadir, it swings ≈55.9 cm below the axle, closest to the transmitter.
- **Dipole arm offset** x_{pc}: -15.7 cm along $+x$ (along the dipole axis).
- **Transverse offset** y_{pc}: -0.5 cm along $+y$, consistent with zero within measurement scatter. This matches the mirror symmetry of the dipole and balun mounting.

The total 3-D lever arm is $\|\mathbf{r}_{\mathrm{pc}}\| = 57.7$ cm, with a mean fundamental lever arm of 57.5 ± 8.7 cm (median 55.9 cm) across sweeps. This directly accounts for the physical placement of the dipole elements, balun, and feed hub on the mast, which sit 40–55 cm from the elevation axle.

Gravitational flexure vs. elevation. Decomposing the elevation path residual into higher Fourier harmonics reveals a persistent 2nd harmonic of amplitude 11.3 ± 2.9 cm, dominated by the cosine term $A_2 \cos(2\mathrm{el}) = +9.6$ cm. At zenith (el = 0°) and nadir (el = 180°), gravity acts along the boresight mast, producing minimal lateral bending. At horizontal pointings (el = ±90°), gravity acts perpendicular to the mast and balun harness, deflecting the cantilevered structure downward toward nadir by ∼11.3 cm relative to the rigid circle.

Platform stability. During continuous motor-driven sweeps, the residual path rms after removing the 3-D lever arm and 2nd-harmonic flexure is 7.0 cm. In stationary dwells (Fig. 11c, `corr_20260717_155704Z.h5`), the antenna experiences high-frequency wind wobble of only 0.75 cm rms on < 5 s timescales (4.19 cm rms over the

full 128 s integration, corresponding to 9.8° of phase at 195.3 MHz), alongside a steady thermal/cable drift of +8.1 cm/min.

Horizon model modulation vs. elevation. The 55.9 cm phase-center offset from the elevation axle shifts the observer's position through which the terrain skyline is seen. For a skyline ridge at horizontal range $r(\phi)$ and elevation angle $\theta_0(\phi)$, the first-order horizon elevation change is $\Delta\theta(\phi; \mathrm{el}) \approx -[\Delta U(\mathrm{el})/r(\phi)] \cos^2 \theta_0(\phi)$. On the current DEM (`derived/dem/v0004`, the DEM the `v0005` horizon profiles use), the skyline ridges stand at ranges from 158.8 m to 7.1 km (median 370.2 m). As the antenna rotates from nadir (calibrations) to zenith (science observations), the phase center rises by $2z_{\mathrm{pc}} \approx 1.11$ m, lowering the apparent horizon across the entire sky by a median of 0.115° (6.9 arcmin) and up to 0.390° (23.4 arcmin) on the closest ridge. Relative to the nominal axle position (`curation/horizon_profiles_v0005`, geometry `v0005`), zenith pointing lowers the horizon by −0.046° (median; worst −0.332°), while nadir pointing raises it by +0.069° (worst +0.220°). At a typical solar elevation rate of ≈ 0.15°/min, this modulation shifts the predicted geometric sunrise and sunset ridge crossings by up to 2.2 min. Integrating the Global Sky Model (GSM2008) through the HFSS beam, held at zenith, with the terrain masked at the axle, zenith and nadir phase-center heights (DEM `v0004` from the `v0005` antenna; ground at 290 K; `validate_horizon_gsm.py`), this modulation changes the predicted antenna temperature at 07-17 06:00 UTC by +2.0 K from the axle to zenith (+4.9 K zenith to nadir; +3.6 to +9.7 K over a sidereal day) at 50 MHz, +16 mK (+47 mK) at 150 MHz, and −0.14 K zenith to nadir at 200 MHz, where the ground is warmer than the sky. Only the mask changes between the three cases, so this isolates the horizon shift from the beam's own rotation.

9 The LIDAR-code fixes

The LIDAR terrain constraint predicted LIDAR ranges from the DEM with three defects, all fixed on 2026-09-18 in the since-retired `terrain` repository. The corrected constraint was first published as `marjum-2026-07/derived/lidar_constraint/v0001`, at the historical `v0001` antenna position. Memo 002 now reports `lidar_constraint/v0003`, the same code at the `v0005` antenna; each version's manifest records its code and inputs.

1. **Sign** (`319f3e9`): the ray bearing is $\Delta_{\mathrm{az}} - \mathrm{az}$, not $\mathrm{az} + \Delta_{\mathrm{az}}$, because bearing decreases as az increases.
2. **Mount** (`4b63b01`): the rays were computed with the alt-az mount. The two mounts agree only at $\mathrm{az} = 0$, which is why the old code still reproduced the dwell plateau. The fix uses Eq. (1) and also removes a double shift of the DEM origin.
3. **Retracted constant**: a nadir elevation of 92.25°, already retracted, was still live. The corrected mount puts nadir at $\mathrm{el} = +90$ by construction and replaces both competing nadir constants by a fitted elevation offset δ (also `4b63b01`).

Later commits fix Δ_{az} at the GPS value 37.836° instead of fitting it (`e153ba4`), cite the free-Δ_{az} scan instead of rerunning it (`371670e`), check that scan's provenance (`0acb255`), and stop a double-anchoring of the DEM cache (`6205128`).

10 Known defects in pointing table v5

- **`EL_POST_FAILURE` never clears.** It latches at the first `EL_STUCK` detection, on 07-14 14:58 (the elevation drive's set screw was replaced on 07-16; memo 001), and stays set on every later row: 57% of the table and all of the raster (Table 2). Do not use it to select scanning data until it is rebuilt to reset after a repair; use `EL_STUCK` and the time of §6.
- **`az_motor_pot_offset_deg`** holds pot−motor, the opposite of its name.
- **The elevation wraps near ±180°** on 314 rows, and the wrap glitches set `UNCOMMANDED_MOTION` on 2,750 rows where the box did not move. Pointing table v6 fixes both (§7.1).
- **The elevation may carry a twice-per-turn error** of 0.55° from the IMU's axes across the axle, possibly up to 0.9°. v6 carries it too, uncorrected, as a systematic (§7).

11 Open questions

- The azimuth handedness rests on the firmware passing `imu_az`'s axes through unchanged, checked on `pico-firmware`'s public `main`, not the build that flew (§5.1). A bench check of the shaft's turn for +motor az would close it.
- With a left-handed azimuth, the 10.3° between the polarization's arm 0 and the photographed rx6 axis. Re-measuring the feed axes in IMG_2203/2204, and tracing which feed quadrant pair the display's "rx6" output drives (field notes, the cabling), would test both readings.
- Why the platform turned less than commanded on 07-13 to 07-15 (`imu_az` phase change 0.33 to 1.20 per degree of motor az, 5–95%; potentiometer down to +0.50 per commanded degree), when on 07-17 it followed the motor to 4.2° rms.
- Absolute azimuth outside the raster. Any epoch with the transmitter on and the antenna rolling would give it by the method of §5; the 07-16 transmitter-on era is a stare and cannot.
- Whether the platform follows the motor in the earlier eras, where the potentiometer and the commanded azimuth disagree most (95th percentile ∼30° in the ∼2 m and ∼91 m eras).
- The IMU elevation zero: the LIDAR route gives −2.3°, the beam fit +0.05° with the transmitter held at its `v0005` value. A measurement of the LIDAR's angle to the boresight would separate them.
- The tether anchor's position, needed to put the 87.5 m and 91 m heights on a datum (memo 002).
- The BNO08x's accelerometer convention (specific force or gravity), which would fix the tilt's sign independently of the radio data, and the accelerometer's bias along the axle.
- The cause of the late-raster under-travel (§6), and whether it led to the 07-18 failure.

Revision history

2026-10-01 First version. Replaces legacy MEMO-002 (pointing table v0) and MEMO-013 (fusion logic), and the mount, elevation and LIDAR-code parts of MEMO-012. New in this memo: the polarization tests of the mount, of commanded azimuth, and of the azimuth zero; the exact raster structure from the motor record; the elevation failure time from the data. Same day, after review: fits with the transmitter's arms held at their photographed bearings favour the right-handed azimuth (zero ≈ −15°) and a swapped arm label over a left-handed azimuth, reversing the first draft's lean.

2026-10-15 Added the late-raster elevation under-travel (§6): from leg 45 the box falls behind the elevation drive on up-legs, inside drive angle 90–150°, until it lags by 28.5°. The table's IMU elevation is unaffected; `motor_el_deg` is not a pointing there. Moved to pointing table v3, which fixes three defects this memo listed for v2.0: rows without a height era or phase (labelled by time interval, now by source file); `EL_SOLUTION_GLITCH` rows left `ok`, and the flag firing on passes through nadir; and the uncommitted generator with its pointer to a notebook in no commit.

Added the current-monitor comparison: the drive remains energised during the under-travel, but the six analysed frozen post-re-raise legs have no measurable running-scale draw. The background is approximately steady, downward-sweep draw declines gradually, and current alone does not exclude supply sag. Named the load thermistor and separated the stillness-detector boundary from the loss-of-response interval. All current numbers are regenerated by the supplied validator and checked against the reviewed reduction.

2026-10-23 Added cross-correlation phase analysis of the 07-17 raster sweeps (§8, Fig. 11). Measured the 3-D antenna phase center lever arm from the elevation axle ($z_{\mathrm{pc}} = +55.6 \pm 6.6$ cm along boresight, $x_{\mathrm{pc}} = -15.7$ cm along dipole arm, $y_{\mathrm{pc}} = -0.5$ cm transverse; total $\|\mathbf{r}_{\mathrm{pc}}\| = 57.7$ cm) and detected 11.3 cm of 2nd-harmonic gravitational flexure vs. elevation. Stationary platform wobble is 0.75 cm rms on < 5 s timescales. All numbers and figures regenerate via `validate_phase_center.py`.

2026-10-25 A direct test of the azimuth handedness on the box's accelerometers (§5, Fig. 3) favours a left-handed azimuth. This overturns the 2026-10-01 inference that a transmitter arm label was swapped: with a left-handed azimuth the labels stand, and a 10.3° arm gap is open. The table and `eigsep_base` keep the

right-handed convention pending a decision. Products regenerated on pointing table v4; no other quoted number moved.

2026-10-28 The azimuth is adopted as left-handed about the boresight, from memo 002's measurement, which moves here (§5.1): `imu_az` on the azimuth shaft, calibrated with its 07-09 elevation revolution, shows +motor az turning the shaft clockwise from above; the beam puts the boresight up at el = 0; two photographs show the dipole turning that way. Pointing table v5 negates the motor and potentiometer readings so that Eq. (1) holds as written, and the table's zero on the raster becomes +15.8°. The reference beam is now `v0017`, fitted on that table. The phase-centre component along the dipole arm changes sign, to −15.7 cm. The earlier accelerometer test of this memo, which measured the phase about the circle's own normal, is replaced by memo 002's calibrated one.

2026-11-04 Added the elevation axle's tilt from the box's IMU (§7). The mount gains the axle's tilt (Eq. 2). Gravity's component along the axle measures it per sample: +5.92° over the 07-17 raster, +0.18° over `fit_v0025`'s 362 night bins and +2.01° after the re-raise, so the beam fit's single +4.74° belongs to the raster alone. The IMU sensor is turned 3.85° in the box, which the firmware's calibration had already measured and the table's elevation already removes. The tilt's sign is taken from the raster beam fit. On the night a sky fit rejects the raster's tilt (held-out χ^2 139.8 at +4.74° against 27.2 at the IMU's and 27.4 at $\tau = 0$) and is consistent with the IMU's value and with level. The night's tilt is quoted as $+0.18 \pm 0.6°$, an allowance for its wander during the night's hold, whose cause is not known; two field sweeps fix the axis to 0.15°. The antenna is bolted to the mount on the box, so the axle tilt is the IMU's. Beam v0018 (memo 005) shows that the raster's fitted tilt predicts held-out data no better than the IMU's, so the raster cannot separate a change of tilt from a small change of the beam's shape. The raster's tilt dips for seven minutes, 20:54–21:01 UTC. Describes pointing table v6, which adds the tilt per sample, fixes an elevation wrap near ±180° on 314 rows, and changes the quality rule: rows not quasi-static or with the elevation undefined become suspect, the corrected wrap rows are no longer suspect for the glitch alone, and uncommanded motion is detected on the accelerometer's elevation. The raster gains 60 `ok` rows facing the transmitter that the beam fit had dropped. The night's waterfall of data, model and residual is now cited from memo 008, Fig. 4; this memo quotes its residuals. The rest of the memo stays on pointing table v5 until v6 is released.

2026-11-06 The IMU tilt analysis now removes the firmware calibration's accelerometer bias from every reading before the tilt. The axle's direction, the raster's tilt, the axle models and the raster's dip change at most in their last quoted digit; the raster keeps 6,517 quasi-static bins. The night's tilt is +0.18° (was +0.19°), and the era means, Table 5 and Fig. 9 follow. The night fit ran with +0.19° and is not rerun: its scan (Fig. 10) and residuals are as it ran, and the text says so (§7). New: the calibration's bias is tested on the raster (Fig. 8). The raster's own fit differs from it by $1.62 \times 10^{-3}\, g$ and changes within the hour, and the readings' magnitude carries a twice-per-turn term of $9.6 \times 10^{-3}\, g$ that the calibration's model does not have; the firmware's bias is kept. That term is fitted as the sensor's two axes across the axle, 91.08° apart, which would put a twice-per-turn error of 0.55° (possibly up to 0.9°) in the table's elevation; at rest only a selection of holds made after seeing the data shows it. The error is carried as a systematic and not corrected. The raster's rotation adds at most $0.43 \times 10^{-3}\, g$ as a centripetal term. The analyses behind this section are named by what they are, and their frozen inputs carry neutral provenance.

A Reproducing this memo

Run `python make_products.py` in this directory, then `pdflatex memo-003.tex` twice. The script reads the inputs below, checks the SHA-256 of the one that is not tracked in git (the beam product's per-sample tooth amplitudes), and writes every figure in `figures/` and every table and quoted number in `tables/`. The tooth amplitudes are background-subtracted transmitter comb power from `empirical_raster_v0017`; they do not depend on pointing, and only their arm-0/arm-1 ratio is used; the azimuths stored beside them are pointing table v5's. The beam fit's geometry corrections come from that product's `provenance.json`, relative to geometry `v0005`.

The current comparison is a *diagnostic* reduction of 72 correlator files. Run `python validate_current.py --raw-root /path/to/corr/files` before `make_products.py` to rebuild `validate_current.json` (`make_products.py --only current` then rereads only it); an existing result is checked against its

source and input hashes before its numbers become macros. The `inputs/current_selection.parquet` and `inputs/current_legs.csv` freeze the reviewed row/leg selections and refit IMU coordinate from Christian Hellum Bye's analysis of the raster record and of the drive current (not shared); they contain no current or temperature measurements. This IMU coordinate defines the current diagnostic's matched-angle window, not a replacement for the v5 pointing table. The validator reads all telemetry anew, uses `time_best` and disables index caching; it records the input-file hashes and checks every quoted current and temperature quantity against the canonical reduction.

The phase-center lever arm and flexure analysis (§8) is regenerated by `python validate_phase_center.py`, which reads the 07-17 raster files and writes `figures/phase_center.pdf` and `validate_phase_center.json`. Both it and `validate_horizon_gsm.py` import the skyline ray-marcher from a clean `data-analysis` checkout named by `$EIGSEP_DATA_ANALYSIS` (or `--data-analysis`), refuse a modified one, and record its commit; they ran at `data-analysis` `e0ceeb3` and `e0ceeb3`. Its GSM antenna-temperature estimate is regenerated by `python validate_horizon_gsm.py`, which reads `validate_phase_center.json`, the `v0004` and `v0005` horizon profiles' antenna positions, DEM `v0004` and the HFSS bowtie beam (`hfss_beam_maps/bowtie_beam.npz` in that checkout), computes GSM2008 with `pygdsm`, and writes `validate_horizon_gsm.json` with its inputs' SHA-256. It also reruns the calculation from the `v0004` antenna, as a check that the script reproduces the earlier unscripted estimate (+2.57 K and +5.32 K at 50 MHz, against +2.57 and +5.32 K quoted before), and the text quotes only the `v0005` case.

The handedness (§5.1) is regenerated by `python validate_az_handedness.py` (raw files under `$EIGSEP_RAW_ROOT/data`; about 15 minutes with 5 workers), which reads `imu_az`'s metadata and calibration, fits the 07-17 raster with each of the four combinations, and writes `validate_az_handedness.json` and `.npz`; it ran at commit `320e242`. `validate_handedness.py` checks the readings' sense (potentiometer against motor, the potentiometer calibrations, and the table's az against motor steps) and writes `validate_handedness.json`; it ran at commit `01723e8`. The photographs' predictions (Fig. 4) use memo 002's camera model, imported from its `make_products.py`, and the geometry release's camera poses; the images' SHA-256 are checked against the release.

Running rows are interior rows of two-sided count changes with the other axis parked; parked rows have neither axis changing on either side. Each running row's background is the mean of nearby parked samples within five minutes, widened until at least two exist. Per-leg draw is the mean above that reference; drift and onset tests use whole legs, an early trend and its extrapolation uncertainty, with a lag-one correlation correction. The IMU-window contrast compares the late up-legs with the same window on earlier up-legs. Frozen-leg draws are pooled across the six selected legs; the comparison running leg is a continuous sweep that morning. Temperatures use the reviewed validity cuts and per-leg medians as onset markers. The monitored rail's exact wiring and load mix remain undocumented; no voltage, torque or motor-temperature inference is made. The current generator ran at memo source `06bd1be`.

The axle-tilt section (§7) is made by `python make_products.py --only tilt` (also part of a full run). Most of its numbers come from Christian Hellum Bye's IMU tilt analysis (not shared; source commit `bb6d90e`), which reads `raw_binned_v0006`, the beam product `v0017`, `fit_v0025` and the night fit, and writes them to `tilt_numbers.json`. The four figures are made here, from its per-bin tilts and readings, `tau_box.npz`: the axle figure (Fig. 7) and its held-out numbers by `tilt_axis.py`, which refits the analysis's three models on the raster and checks that they reproduce its results (`imu_tilt_results.json`); the test of the calibration's bias (Fig. 8) by the same script, whose circle fit repeats the firmware's (`picohost.imu_geometry.fit_accel_sphere_coplanar`, checked against it once, with unit weights) and adds the twice-per-turn term, with the IMU elevation analysis's fit of that term (its `results.json`, as `imu_elevation_results.json`, source commit `dad46d0`); the tilt over phase C (Fig. 9) and the night's scan (Fig. 10) by `tilt_eras.py`, with `fit_v0025`'s fitted bins (checked against the analysis's). The scan is refitted from the night fit's products (its `results.json` from the run after its loss prior was rebuilt from the parts): its binned data with the wind cut's mask (`night_fit_data_v0025.npz`) and its sky and ground columns at each tilt of the scan (`night_fit_run_<set>.npz`), checked against its `results.json` (as `night_fit_results.json`); its jackknife follows the IMU tilt analysis's method, and the per-bin tilt's score uses that analysis's column set (`imu_tilt_run_taubinc0.npz`). The numbers on the night's data, model and residual are refitted by `tilt_dmr.py` from the same products (on the bins the wind cut keeps, at the IMU's and the beam fit's tilt), and checked against the night fit's `results.json`, so this memo quotes the fits shown in memo 008, Fig. 4. The per-era tilt and the description of v6 come from pointing table v6's validation (`validation_v6.json`) and the comparison of its `cb6e763` build with v5 (`compare_first_build.json`, of which only the v5 and `cb6e763` counts

are used). Those counts hold for v6 as built (`v6+4af5ce9`), which differs from `cb6e763` only in `tilt_sigma_deg` (`compare_sigma_build.json`). All are copied into `inputs/tilt/`, and the script checks their SHA-256 before using them. The night fit ran with the tilt held at +0.19°; the script checks that value against the IMU tilt analysis's night mean on the readings before the bias was removed (`imu_tilt_results.json`) and draws the scan's star there (§7).

input	SHA-256 (first 16)
`marjum-2026-07/curation/pointing_table.parquet`	`0f03c6de2924dd13`
`marjum-2026-07/imgs/fits/v0005_marjum_geometry/shared.json`	`2ec618bff168c1b0`
`marjum-2026-07/derived/beam/empirical_raster_v0017/dpss/diagnostics.npz`	`d3f0dd9e3abc7a38`
`marjum-2026-07/derived/beam/empirical_raster_v0017/provenance.json`	`ca77de003dfa1206`
`marjum-2026-07/derived/tx_feed_orientation/v0003/summary.json`	`3226392671618ba8`
`marjum-2026-07/derived/lidar_constraint/v0003/summary.json`	`30a90a505c043ed0`
`memos/memo-002-marjum-2026-07-geometry/tables/summary.json`	`7b1f0684903e71c5`
`memos/memo-003-marjum-2026-07-pointing/validate_current.json`	`b852eb4b1b2eb3a5`
`memos/memo-003-marjum-2026-07-pointing/inputs/current_manifest.json`	`bea7a6ae7e432616`
`memos/memo-003-marjum-2026-07-pointing/inputs/current_selection.parquet`	`c86260c4e4ffebff`
`memos/memo-003-marjum-2026-07-pointing/inputs/current_legs.csv`	`b23fab0fa935c2bb`
`memos/memo-003-marjum-2026-07-pointing/validate_phase_center.json`	`f692c799dd4a9d58`
`memos/memo-003-marjum-2026-07-pointing/validate_handedness.json`	`cddee6e898450ac1`
`memos/memo-003-marjum-2026-07-pointing/validate_az_handedness.json`	`2a6e2cda87a0c2ba`
`memos/memo-003-marjum-2026-07-pointing/validate_az_handedness.npz`	`526b037f002a8f89`
`memos/memo-003-marjum-2026-07-pointing/validate_horizon_gsm.json`	`060848592cdd117b`
`marjum-2026-07/imgs/fits/v0005_marjum_geometry/cameras.jsonl`	`fbbc074ea98312e3`
`marjum-2026-07/imgs/fits/v0005_marjum_geometry/manifest.json`	`7028df7d56f01934`
`memos/memo-002-marjum-2026-07-geometry/make_products.py`	`155f6dc089703bfc`
`marjum-2026-07/imgs/IMG_2217.HEIC`	`843581e8353ae2ad`
`marjum-2026-07/imgs/IMG_2226.HEIC`	`29777849b495686e`
pointing table version	`v5+f07f038`
eigsep (memo, campaign) commit	`6edcf2a`
eigsep_base commit	`840c10b`
generated	2026-11-06 16:26 UTC

axle-tilt input (§7)	SHA-256 (first 16)
memos/memo-003-marjum-2026-07-pointing/inputs/tilt/beam_v0018_compare.json	8f797c63290080ea
memos/memo-003-marjum-2026-07-pointing/inputs/tilt/beam_v0018_window.json	d6a341d84c9fe622
memos/memo-003-marjum-2026-07-pointing/inputs/tilt/compare_first_build.json	c0472562dc2f03f1
memos/memo-003-marjum-2026-07-pointing/inputs/tilt/compare_sigma_build.json	eb649587f81e8f72
memos/memo-003-marjum-2026-07-pointing/inputs/tilt/imu_elevation_results.json	756dd55b0e0b7657
memos/memo-003-marjum-2026-07-pointing/inputs/tilt/imu_tilt_results.json	571267006a8d30ad
memos/memo-003-marjum-2026-07-pointing/inputs/tilt/imu_tilt_run_taubinc0.npz	240c2edaf16730d0
memos/memo-003-marjum-2026-07-pointing/inputs/tilt/night_fit_data_v0025.npz	740d7aa207d90689
memos/memo-003-marjum-2026-07-pointing/inputs/tilt/night_fit_results.json	9953b170cd0bb4fe
memos/memo-003-marjum-2026-07-pointing/inputs/tilt/night_fit_run_nominal.npz	353e36545da1688f
memos/memo-003-marjum-2026-07-pointing/inputs/tilt/night_fit_run_tau1p0.npz	ed647aab51ed8cfb
memos/memo-003-marjum-2026-07-pointing/inputs/tilt/night_fit_run_tau2p5.npz	3d1e4a7946a1d358
memos/memo-003-marjum-2026-07-pointing/inputs/tilt/night_fit_run_tau3p5.npz	5feea17ab27e2f1a
memos/memo-003-marjum-2026-07-pointing/inputs/tilt/night_fit_run_taum4p74.npz	8ad281527145a7aa
memos/memo-003-marjum-2026-07-pointing/inputs/tilt/night_fit_run_tauv0017.npz	65b217c7fd9ec25a
memos/memo-003-marjum-2026-07-pointing/inputs/tilt/night_fit_run_tilt0.npz	3324d4a84c07780b
memos/memo-003-marjum-2026-07-pointing/inputs/tilt/tau_box.npz	b9744badd8f58d6b
memos/memo-003-marjum-2026-07-pointing/inputs/tilt/tilt_numbers.json	3a0100f355999cbf
memos/memo-003-marjum-2026-07-pointing/inputs/tilt/validation_v6.json	a7d7b40f6dbb3a0d
IMU tilt analysis, source commit (not shared)	bb6d90eb2a
pointing table version	v6+4af5ce9
eigsep (memo) commit, tilt products	7443145
generated	2026-11-07 00:07 UTC